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Improved compression inference algorithm for reliability analysis of complex multistate satellite system based on multilevel Bayesian network

Xiaohu Zheng[a], Wen Yao[b,*], Yingchun Xu[a], Xianqi Chen[a]

[a] College of Aerospace Science and Engineering, National University of Defense Technology, No. 109 Deya Road, Kaifu District, Changsha, Hunan 410073, China
[b] National Innovation Institute of Defense Technology, Chinese Academy of Military Science, No.53, East Main Street, Fengtai District, Beijing 100071, China

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ABSTRACT

Bayesian Network (BN) is a powerful tool for analyzing system reliability. However, for the complex multistate satellite system, the state combination explosion makes the reliability analysis based on BN computationally unaffordable. To solve this problem, compression algorithm and inference algorithm have been proposed. The original compression algorithm classifies all the phrases with same composition into the same type. The original compression algorithm classifies all the phrases with same composition into the same type. However, cases tested in this paper show that the results of inference can be incorrect when some node probability tables have special formulations. To solve the above defects, the compression algorithm and sequential inference algorithm are improved in this paper. Based on improved compression algorithm and improved inference algorithm, the improved compression inference algorithm (ICIA) is formed and extended to the application of multistate nodes with independent binary parent nodes. Besides, the multilevel BN (MBN) models are used to construct the BN models of complex multistate satellite systems so as to further improve inference efficiency. Cases tested in this paper show that the proposed algorithms are more efficient than the Bayes Net Toolbox and the AgenaRisk software for the reliability analysis of complex multistate satellite systems.

1. Introduction

In system engineering, the existence of uncertainty has a great impact on system performance. Currently, the main system uncertainty analysis methods include Bayesian network (BN) [1, 2], monte-carlo simulation [3, 4], approximation method for mixed aleatory and epistemic uncertainties [5, 6], Bayesian melding method [7, 8], etc. Among these methods, BN has powerful uncertainty reasoning ability and has been applied to reliability analysis [4, 9], safety analysis [10, 11], fault analysis [12, 13], domino effect analysis [14, 15], risk analysis [16–18], etc. In particular, system reliability analysis is an important issue and is becoming highly demonding, especially for the high-value complex systems, e.g. satellite system [19].

As the complexity of the system increases, reliability analysis of complex systems is becoming more challenging. For complex systems, Adedigba et al. [11, 20]. analyzed the safety of a non-sequential barrier-based process accident model by reliability data based on BN and Leaky Noisy-OR gate model. In the researches of Sunday Adedigba et al., the BN models of complex systems emphasize the existence of nonlinear interactions between nodes. What's more, all nodes in BN only have two states, i.e. total failure and total normal. However, different degrees of damage have different influences on system performance. For example, the satellite system has attitude determination and control subsystem (ADCS), power subsystem and some other subsystems. Assuming ADCS partially fails and cannot control its attitude following the required high accuracy, but it can be turned into rough control mode and continues the sensing functions, which means the satellite is not completely scrapped. In this situation, the ignored partial failures of systems may lead to the inaccurate results of systems reliability analysis. Therefore, multistate systems have been paid much attention to and some BN-based methods have been proposed for multistate systems reliability analysis over the past several years. Based on BN, Mi et al. [21, 22] proposed new methods to analyze the reliability of complex multi-state system with common cause failure. Besides, based on interval fuzzy Bayesian network, Mi et al. [23] prooposed a method for reliability modeling and analysis of complex multistate system with uncertainties and failure dependencies. However, as the functions of complex systems increase, the number of system units will rise. Apparently, the number of state combinations of units will also increase exponentially. Therefore, one of the major challenges for the BN-based reliability analysis methods for complex systems are the construction of node probability table (NPT) in the BN model and the inference of the BN model.

* Corresponding author.
E-mail address: wendy0782@126.com (W. Yao).

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Nomenclature

n	the parent node numbers of a child node
C_i	the i th parent node of a child node
L	the row number of a child node's NPT, $L = 1, 2, ..., 2^n$
s_i^L	the state of parent node C_i in the L th row of NPT
r^j	the numerical value that makes up the run in the j th row of compressed sequence
L_r^j	the length of the run in the j th row of compressed sequence
p^j	the phrase number in the accompanying dictionary, where the phrase is in the j th row of compressed sequence
n_p^j	the repeated instances numbers of the phrase in sequence, where the phrase is in the j th row of compressed sequence
v_1^j	the first numerical value that makes up the phrase in the j th row of compressed sequence
v_2^j	the second numerical value that makes up the phrase in the j th row of compressed sequence
L_p^j	the length of the phrase in the j th row of compressed sequence
λ_i	the intermediate factor which is created after eliminating the parent node C_i
$c\lambda_i$	the compressed λ_i
d_i	the accompanying dictionary of $c\lambda_i$
$c\lambda_i^j$	the j th row of $c\lambda_i$
d_i^j	the j th row of d_i
r_i^j	the numerical value that makes up the run in the j th row of $c\lambda_i$
$L_{r_i}^j$	the length of the run in the j th row of $c\lambda_i$
p_i^j	the phrase number in the accompanying dictionary, where the phrase is in the j th row of $c\lambda_i$
$n_{p_i}^j$	the repeated instances numbers of the phrase in λ_i, where the phrase is in the j th row of $c\lambda_i$
$v_{1_i}^j$	the first numerical value that makes up the phrase in the j th row of $c\lambda_i$
$v_{2_i}^j$	the second numerical value that makes up the phrase in the j th row of $c\lambda_i$
S_i^j	run or phrase start row number in λ_i, where the run or the phrase is in the j th row of $c\lambda_i$
S^{now}	the current querying phrase start row number in uncompressed objective sequence
S^{exist}	the existing phrase of the accompanying dictionary start row number in uncompressed objective sequence
RP_i^j	If the j th row of $c\lambda_i$ is a run, RP_i^j is the run start row number in λ_i. If the j th row of $c\lambda_i$ is a phrase, RP_i^j is the set of all the same type phrase start row numbers in λ_i.
m_i	the row numbers of $c\lambda_i$
RP_i	the set of RP_i^j, i.e. $\{RP_i^j \mid j = 1, 2, ..., m_i\}$ a set that is composed by the start row numbers of all run and phrase of $c\lambda_i$
L_{RP}	the element numbers of RP_{i+1}^j for a phrase
$J_{i+1}^j(1, i_{RP})$	the i_{RP} th element of J_{i+1}^j, $i_{RP} = 1, 2, ..., L_{RP}$
$RP_{i+1}^j(1, i_{RP})$	the i_{RP} th element of RP_{i+1}^j
R_{i+1}^j	the remainder after processing row j of $c\lambda_{i+1}$
R^{all}	the set of R_{i+1}^j
$R_{J_{i+1}^j(1,i_{RP})}^{all}$	the $J_{i+1}^j(1, i_{RP})$ th element of R^{all}
$R1_{i+1}^j$	the start row number of the first equivalent run
$R2_{i+1}^j$	the start row number of the second equivalent run
RP_{i+1}^{update}	the updated RP_{i+1}
S_{i+1}^{update}	the updated S_{i+1}^{all}
$c\lambda_i^{new}$	the newly obtained compressed intermediate factor which is created after eliminating the parent node C_i
λ_i^{new}	the decompressed and reorganized NPT of $c\lambda_i^{new}$
$c\lambda_i^{new-j_1}$	the j_1 th row of $c\lambda_i^{new}$
d_i^{new}	the newly obtained accompanying dictionary of $c\lambda_i^{new}$
$d_i^{new-j_1}$	the j_1 th row of d_i^{new}
$r_i^{new-j_1}$	the numerical value that makes up the run in the j_1 th row of $c\lambda_i^{new}$
$L_{r_i}^{new-j_1}$	the length of the run in the j_1 th row of $c\lambda_i^{new}$
$p_i^{new-j_1}$	the phrase number in the accompanying dictionary, where the phrase is in the j_1 th row of $c\lambda_i^{new}$
$n_{p_i}^{new-j_1}$	the repeated instances numbers of the phrase in λ_i^{new}, where the phrase is in the j_1 th row of $c\lambda_i^{new}$
$v_{1_i}^{new-j_1}$	the first numerical value that makes up the phrase in the j_1 th row of $c\lambda_i^{new}$
$v_{2_i}^{new-j_1}$	the second numerical value that makes up the phrase in the j_1 th row of $c\lambda_i^{new}$
$L_{p_i}^{new-j_1}$	the length of the run in the j_1 th row of $c\lambda_i^{new}$
$L_{c\lambda}$	the length of RP_{i+1}^{update}, $j_2 = 1, 2, ..., L_{c\lambda}$
$\mathbf{V}_\lambda$	a $L_{p_i}^{new-j_2} \times 1$ matrix
L_S	the length of S_{i+1}^{update}
$\mathbf{V}_2$	a matrix whose size and all elements are $L_p \times 1$ and $v_{2_i}^{new-j_2}$ respectively
$\mathbf{V}_\lambda'$	a 2×1 cell array whose the first cell is $v_{1_i}^{new-j_2}$ and the second cell is $\mathbf{V}_2$
Po_λ	For a run, Po_λ is a scalar and $\mathbf{A}_\lambda\{Po_\lambda, 1\} = \mathbf{V}_\lambda$ represents that the Po_λ th cell of $\mathbf{A}_\lambda$ is $\mathbf{V}_\lambda$. While for a phrase, Po_λ is an array and $Po_{k_1}^\lambda$ is the k_1 th element of Po_λ.
N_{unit}	the binary independent units numbers of multistate satellite system Sys, $r = 1, 2, ..., N_{unit}$
N_{state}	the state numbers of multistate satellite system Sys
U_{prq}	the rth node of the pth level
N	the level numbers of MBN, $p = 1, 2, ..., N$
J_{i+1}^j	For a run, J_{i+1}^j represents the position of RP_{i+1}^j in S_{i+1}^{all}. For a phrase, J_{i+1}^j represents the set of the positions of all the elements of RP_{i+1}^j in S_{i+1}^{all}.

To solve the problem of state combinations increasing exponentially, Tien and Kiureghian [24] proposed compression algorithm and inference algorithm for reliability assessment of infrastructure systems. By compressing the system column of the NPT, compression algorithm can reduce the memory storage requirements in constructing the BN model significantly. Based on the compressed system column of the NPT, inference algorithm can perform inference efficiently. However, the proposed algorithms by Tien and Kiureghian can only be applied to binary systems. Besides, compression algorithm and inference algorithm have some defects, e.g., the results of inference will be incorrect and sometimes the inference cannot be completed for some special NPTs which will be described in detail in Section 2.3. Extending the binary system to multistate system, Tong and Tien [25] proposed new algorithms to construct the BN model for the multistate systems reliability analysis, but the number of states of components and system must be equal. For example, the authors supposed that both components and system had five states in the studied systems.

For some complex satellite systems, multistate systems with binary units or subsystems usually can reflect an accurate relationship between systems and its units or subsystems for reliability analyses. Therefore, new algorithms are needed for the BN modeling and reliability analysis of complex multistate satellite systems with binary units or subsystems. The assumption that the root nodes or bottom level subsystems have only binary states is based on the following considerations. In practical engineering, many basic elements, especially the electronic components, e.g. resistor, electromagnetic relay, reaction wheel, etc., are typical binary state units. By proper combination of these units, the composed subsystem or system (the child node of the combination of

the parent nodes) has multiple states. This is very popular in practical engineering and accordingly the proposed method in this paper can cover a large range of the practical application scenarios.

The rest of this paper is structured as follows. In Section 2, a brief background on compression algorithm and inference algorithm is provided, wherein the format of node probability table and the defects of the compression algorithm and inference algorithm are also presented. To tackle the defects of original compression algorithm and inference algorithm, an improved compression algorithm and an improved inference algorithm are developed to ensure the probability inference results of any binary nodes are correct in Section 3. In Section 4, an novel improved compression inference algorithm (ICIA) is proposed to extend the compression technique to the probability inference of multistate nodes. Besides, a multilevel BN (MBN) reliability inference algorithm is also developed to draw inferences of the whole MBN models of complex multistate satellite systems in this section. Three numerical cases are studied to validate the proposed algorithms in Section 5. Finally, the conclusions are presented in Section 6.

2. Background

2.1. Format of NPT

For a root node, the NPT is a marginal probability distribution. For a child node, the NPT is a conditional probability distribution which describes the relationship between the states of the child node and every state combination of the parent nodes. In this paper, the format of NPT is shown as follow for a child node.

Suppose that a child node Ch has n parent nodes as shown in Fig. 1, where each parent node is denoted as $C_i (i = 1, 2, ..., n)$. Both the child node and parent nodes are binary nodes, and "0" means that the node is in state 1 and "1" means that the node is in state 2. Besides, $\Pr(Ch = 0|C_1, C_2, ..., C_n)$ denotes the conditional probability of the child node Ch in state 1, while $\Pr(Ch = 1|C_1, C_2, ..., C_n)$ denotes the conditional probability of the child node Ch in state 2. The pattern of the parent nodes state combinations is constructed according to the rule in [24] as shown in the following.

L is the row number of NPT for Ch, i.e. $L = 1, 2, ..., 2^n$. s_i^L is the state of parent node C_i in the L th row of NPT. The value of s_i^L is calculated according to formula (1)

$$s_i^L = \begin{cases} 0 & \text{if } ceil\left(\frac{L}{2^{n-i}}\right) \in \text{odd} \\ 1 & \text{if } ceil\left(\frac{L}{2^{n-i}}\right) \in \text{even} \end{cases} \quad (1)$$

where $ceil(x)$ is the value of x rounded up to the nearest integer.

For example, suppose the binary child node Ch has 3 binary parent nodes which are in parallel. Because $ceil(1/2^{3-1}) = 1$ is odd, $s_1^1 = 0$, i.e. the state of C_1 in the first row of NPT is 0. Similarly, because $ceil(2/2^{3-1}) = 1$ is odd, $s_1^2 = 0$, i.e. the state of C_1 in the second row of NPT is 0. In analogy with the above calculation process, the rest of the state of each parent node in NPT can be obtained. Then, the value of each row in the column $\Pr(Ch|C_1, C_2, C_3)$ of NPT can be gotten according to the logical relation between the child node Ch and its 3 parent nodes. Thus, the NPT of child node Ch is obtained as shown in Table 1.

2.2. Reliability analysis based on original algorithms

2.2.1. Compression algorithm

The compression algorithm in [24] is used to reduce the memory storage requirement of NPT. The compression algorithm integrates run-length encoding compression technique [26] and Lempel-Ziv encoding compression technique [27]. Because both of these two techniques are lossless techniques, no approximations are made in the process of compressing NPT. The compression algorithm is used to compress the column corresponding to $\Pr(Ch = 1|C_1, C_2, ..., C_n)$ in the NPT of Ch.

A run is defined as consecutive sequence of the same value, e.g. a consecutive sequence of 1 s is a "1" run. A sequence like "0111…1" is a phrase of which the second bit is different from the first bit but is the same with the bits from the third to the last. In the compressed column $\Pr(Ch = 1|C_1, C_2, ..., C_n)$, the j th row is $\{run, r^j, L_r^j\}$ or $\{phrase, p^j, n_p^j\}$, where j is the row number of the compressed column $\Pr(Ch = 1|C_1, C_2, ..., C_n)$, "*run*" or "*phrase*" defines the type of this row, r^j is the value of the run, p^j is the phrase number in the accompanying dictionary, L_r^j is the length of the run and n_p^j is the number of repeated instances of the "phrase" in sequence. In the accompanying dictionary, the p^j th row is $\{p^j, v_1^j, v_2^j, L_p^j\}$, where v_1^j is the first value in the phrase, v_2^j is the second and subsequent values in the phrase and L_p^j is the length of the "phrase".

The compression process is that the column $\Pr(Ch = 1|C_1, C_2, ..., C_n)$ is queried one bit by one bit from the first row to the last row. If the current querying bit is different from the previous bit, e.g. the current querying value "1" following the previous value "0", it means that the current querying bit will be the beginning of a new run or the beginning of a phrase. If the next bit is the same as the current querying bit, i.e. a "1" following the current querying value "1", then it means that the current querying bit is the beginning of a new run. If the next bit is different from the current querying bit, i.e. a "0"; following the current querying value "1", then it means the current querying bit is the beginning of a phrase. The phrase has to be comprised of two elements at least. Once the phrase appears, the accompanying dictionary should be built along with this phrase. If the current querying phrase already exists in the accompanying dictionary, the number of repeated instances of this phrase in sequence will be updated accordingly but the current querying phrase will not be added to the accompanying dictionary.

For example, the binary child node Sy has 3 independent binary parent nodes and the NPT of Sy is shown in Table 2. The compressed column $\Pr(Sy = 1|C_1, C_2, C_3)$ is denoted as $cNPT_1^{Sy}$. The process of compressing the column $\Pr(Sy = 1|C_1, C_2, C_3)$ of Table 2 is shown in the following

For the column $\Pr(Sy = 1|C_1, C_2, C_3)$, the first bit is 0 and the second bit is 1, i.e. the first bit in the sequence is different from the second bit. Besides, the third bit is 0 which is different from the second bit. Thus, the first two bits compose a phrase, i.e. "01". Therefore, $p^1 = 1$, $n_p^1 = 1$, $v_1^1 = 0$, $v_2^1 = 1$ and $L_p^1 = 2$, i.e. the first row of $cNPT_1^{Sy}$ is $\{phrase, 1, 1\}$ and the p^1 th row of the accompanying dictionary of $cNPT_1^{Sy}$ is {1, 0, 1, 2}. Back to the query, the fourth bit is 1 and the fifth bit is 0. Thus, the third bit and the fourth bit make up a phrase, i.e. "01". In particular, the "01" has existed in the accompanying dictionary of $cNPT_1^{Sy}$. Therefore, the n_p^1 is updated, i.e. $n_p^1 = 2$. The values of other variable of this phrase remain unchanged. Back to the query, both the sixth bit and the seventh bit are 1 which is different from the fifth bit. The last bit is 0 which is different from the seventh bit. Thus, the fifth bit, the sixth bit and the seventh bit make up a new phrase, i.e. "011". Therefore, $p^2 = 2$, $n_p^2 = 1$, $v_1^2 = 0$, $v_2^2 = 1$ and $L_p^1 = 3$, i.e. the second row of $cNPT_1^{Sy}$ is $\{phrase, 2, 1\}$ and the p^2 th row of the accompanying dictionary of $cNPT_1^{Sy}$ is {2, 0, 1, 3}. The last bit itself composes a "1"; run. Therefore, $r^3 = 0$, $L_r^3 = 1$, i.e. the third row of $cNPT_1^{Sy}$ is $\{run, 1, 1\}$. To sum up the results of the above

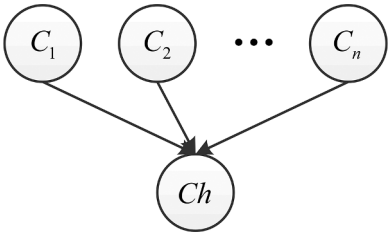


Fig. 1. A Child node Ch with n parent nodes.

Table 1
NPT for the child node *Ch*.

C_1	C_2	C_3	$\Pr(Ch\|C_1, C_2, C_3)$ $Ch = 0$	$Ch = 1$
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

Table 2
NPT of *Sy*.

C_1	C_2	C_3	$\Pr(Sy\|C_1, C_2, C_3)$ $Sy = 0$	$Sy = 1$
0	0	0	1	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

process, $cNPT_1^{Sy}$ and its accompanying dictionary d_{CA}^{Sy} are shown in Tables 3 and 4, respectively.

2.2.2. Inference algorithm

In the variable elimination algorithm (VE) [28], the inference is performed by eliminating the nodes one by one. For example, for the node *Sy* in Table 2, suppose that C_1 and C_2 are the query nodes, i.e. $Q = \{C_1, C_2\}$. Thus, based on VE, the conditional probability $\Pr(Sy|Q)$ is calculated by

$$\begin{aligned}\Pr(Sy|Q) &= \sum_{C_3} \Pr(C_3)\Pr(Sy|C_1, C_2, C_3)\\ &= \Pr(Sy|C_1, C_2)\\ &= \lambda_3\end{aligned} \tag{2}$$

where the $\Pr(C_3)$ is the marginal probability of the parent node C_3 and $\lambda_3 = \Pr(Sy|C_1, C_2)$ is the intermediate factor which is obtained after eliminating C_3.

λ_i represents the intermediate factor which is created after eliminating the parent node C_i. $c\lambda_i$ is the compressed λ_i and $c\lambda_i^j$ is the j th row of $c\lambda_i$. Similar to the rows of the compressed column $\Pr(Ch = 1|C_1, C_2, ..., C_n)$ of NPT, $c\lambda_i^j$ is $\{run, r_i^j, L_{r_i}^j\}$ or $\{phrase, p_i^j, n_{p_i}^j\}$. d_i represents the accompanying dictionary of $c\lambda_i$. d_i^j means the p_i^j th row of d_i and d_i^j is $\{p_i^j, v_{1_i}^j, v_{2_i}^j, L_{p_i}^j\}$. S_i^j is a run or phrase start row number in λ_i, where the run or phrase is the first value of the j th row of $c\lambda_i$. Based on VE, the inference algorithm in [24] performs the inference by using the compressed NPT and its accompanying dictionary for independent binary nodes. In the inference algorithm, the query parent nodes are ordered to extreme left of the NPT of the child node *Ch*. When the parent node C_i is eliminated, according to the parity of S_{i+1}^j, $L_{r_{i+1}}^j$ and $L_{p_{i+1}}^j$, the inference algorithm constructs $c\lambda_i^j$ and d_i^j by using the rules of [24] as shown in Appendix A and Appendix B.

For example, for the node *Sy* in Table 2, suppose $\Pr(C_3 = 0) = 0.1$. Unlike the Eq. (2), based on Tables 3 and 4, the conditional probability $\Pr(Sy|Q)$ is calculated by using the inference algorithm. After eliminating C_3, $c\lambda_3$ and d_3 are constructed by using the rules in Appendix A and Appendix B. For the first row of Table 3, $S_{Sy}^1 = 1 \in$ odd and $L_{p_{Sy}}^1 = 2 \in$ even:

$$\begin{cases} p_3^1 = p_{Sy}^1 = 1\\ n_{p_3}^1 = n_{p_{Sy}}^1 = 2\\ v_{1_3}^1 = \left[v_{1_{Sy}}^1 \times \Pr(C_3 = 0)\right] + \left[v_{2_{Sy}}^1 \times \Pr(C_3 = 1)\right] = 0.9\\ v_{2_3}^1 = v_{2_{Sy}}^1 = 1\\ L_{p_3}^1 = \left(L_{p_{Sy}}^1 - 2\right)/2 + 1 = (2-2)/2 + 1 = 1 \end{cases} \tag{3}$$

Thus, $c\lambda_3^1 = \{phrase, 1, 2\}$ and $d_3^1 = \{1, 0.9, 1, 1\}$. Similar to the calculation of $c\lambda_3^1$ and d_3^1, according to the second row and the third row of Table 3, $c\lambda_3^2 = \{phrase, 2, 1\}$, $d_3^2 = \{2, 0.9, 1, 1\}$ and $c\lambda_3^3 = \{run, 0.1, 1\}$ can be obtained respectively. To sum up the above results, $c\lambda_3$ and d_3 are shown in Tables 5 and 6, respectively. Decompressing Tables 5 and 6, the $\Pr(Sy|Q)$ is gotten as shown in Table 7.

2.2.3. Reliability analysis

Based on compression algorithm and inference algorithm, a method for analyzing binary system reliability was proposed by Tien and Kiureghian [24]. According to the set of minimum cut sets (MCSs) of the system, the NPT of system is compressed by compression algorithm to get the compressed NPT and its accompanying dictionary. After that, the inference of system BN is performed by the variable elimination algorithm based on the compressed NPT and its accompanying dictionary. The analysis process is shown in Fig. 2.

2.3. Defects of compression algorithm and inference algorithm

(1) Defect of compression algorithm

According to compression algorithm, if the current querying phrase has already existed in the accompanying dictionary, the number of repeated instances of this phrase in sequence will be updated accordingly but the current querying phrase will not be added to the accompanying dictionary. However, according to the inference algorithm, in order to ensure the outputs of inference algorithm be correct, the repeated instances of phrase must meet the following two conditions.

a) The parity of the start row number S^{now} of current querying phrase and the start row number S^{exist} of the phrase which is already in the accompanying dictionary must be consistent, i.e. if $S^{exist} \in$ odd, $S^{now} \in$ odd or if $S^{exist} \in$ even, $S^{now} \in$ even.
b) The repeated instances of the phrase must be continuous in the sequence.

For the column $\Pr(Sy = 1|C_1, C_2, C_3)$ of Table 2, the repeated instances of the phrase "01" meet the above two conditions at the same time. No matter which of the two "01" phrases is used to calculate $v_{1_3}^1$, the value of $v_{1_3}^1$ in Eq. (3) is exactly the same. However, suppose that the NPT of the node *Sy* is a special NPT as shown in Table 8 in which the repeated instances of phrase do not meet the first condition.

For Table 8, the compressed column $\Pr(Sy = 1|C_1, C_2, C_3)$ (denoted as $cNPT_{special}^{Sy}$) and its accompanying dictionary $d_{special}^{Sy}$ are shown in Tables 9 and 10, respectively. According to Tables 9 and 10, the value of $v_{1_3}^1$ is calculated by using the first phrase "011" and the second phrase "011", respectively:

For the first phrase "011", $S_{Sy}^1 = 1 \in$ odd and $L_{p_{Sy}}^1 = 3 \in$ odd :

Table 3
Compressed column$\Pr(Sy = 1|C_1, C_2, C_3)$.

run or phrase	r^jorp^j	L_r^j or n_p^j
phrase	1	2
phrase	2	1
run	0	1

Table 4
Accompanying dictionaryd_{CA}^{Sy}.

p^j	v_1^j	v_2^j	L_p^j
1	0	1	2
2	0	1	3

Table 5
Compressed λ_3.

run or phrase	r_3^j or p_3^j	L_{r3}^j or n_{p3}^j
phrase	1	2
phrase	2	1
run	0.1	1

Table 6
Accompany dictionary of $c\lambda_3$.

p_3^j	$v_{1_3}^j$	$v_{2_3}^j$	L_{p3}^j
1	0.9	1	1
2	0.9	1	1

Table 7
Conditional probability Pr($Sy|Q$).

C_1	C_2	Pr($Sy|C_1, C_2$) $Sy = 0$	$Sy = 1$
0	0	0.1	0.9
0	1	0.1	0.9
1	0	0.1	0.9
1	1	0.9	0.1

$$v_{1_3}^1 = \left[v_{1_{Sy}}^1 \times \Pr(C_3 = 0)\right] + \left[v_{2_{Sy}}^1 \times \Pr(C_3 = 1)\right] = 0.9 \quad (4)$$

For the second phrase "011", $S_{Sy}^1 = 4 \in$ even and $L_{P_{Sy}}^1 = 3 \in$ odd :

$$v_{1_{C_3}}^1 = [1 \times \Pr(C_3 = 0)] + \left[v_{1_{Sy}}^1 \times \Pr(C_3 = 1)\right] = 0.1 \quad (5)$$

Apparently, the above two results are different. Similarly, suppose that the column Pr($Sy = 1|C_1, C_2, C_3$) of the NPT of the node Sy is "01101011" of which the repeated instances of phrase "011" do not meet the above two conditions at the same time. Repeating the above calculation process, the value of $v_{1_3}^1$ calculated by using the first phrase

Table 8
Special NPT of Sy.

C_1	C_2	C_3	Pr($Sy|C_1, C_2, C_3$) $Sy = 0$	$Sy = 1$
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	0
1	1	1	0	1

Table 9
Compressed $Sy = 1$ column of Table 8.

run or phrase	r^jorp^j	L_r^j or n_p^j
phrase	1	2
phrase	2	1

Table 10
Accompany dictionary of $cNPT_{special}^{Sy}$.

p	v_1^j	v_2^j	L_p^j
1	0	1	3
2	0	1	2

"011" is also different from what calculated by using the second phrase "011". Therefore, not all the phrases with the same composition can be classified into the same type of phrase.

(2) Defect of inference algorithm

It is noteworthy that both $L_{P_3}^1$ and $L_{P_3}^2$ in Table 6 are equal to 1. However, according to the definition of phrase, the phrase should have 2 bits at least, i.e. $L_{P_i}^j \geq 2$. Therefore, both $v_{1_3}^1$ and $v_{2_3}^1$ in Table 6 do not exist in fact. For example, $c\lambda_3$ as shown in Table 5 is decompressed, the new NPT of Sy can be obtained as shown in Table 7. Then, based on compression algorithm, the compressed column Pr($Sy = T|C_1, C_2$) (denoted as $c\lambda_3'$) is obtained as shown in Table 11 by compressing column Pr($Sy = T|C_1, C_2$) of Table 7.

Apparently, by comparing with $c\lambda_3$ in Table 5, $c\lambda_3'$ in Table 11 only includes the run "0.9 0.9 0.9" and the run "0.1" without any phrase.

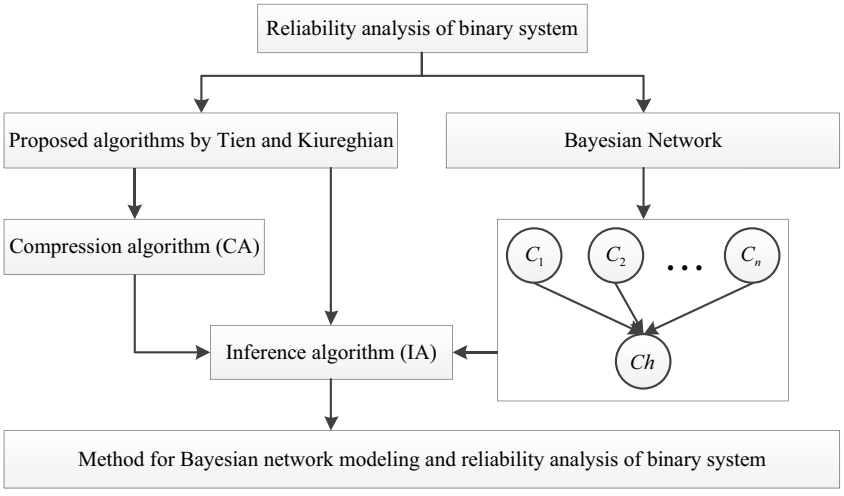


Fig. 2. Method for Bayesian network modeling and reliability analysis of binary system.

Table 11
$c\lambda_3'$ based on compression algorithm.

run or phrase	$r_3^{j'}$ or $p_3^{j'}$	$L_{r3}^{j'}$ or $n_{p3}^{j'}$
run	0.9	3
run	0.1	1

Therefore, whenever a parent node is eliminated in the process of inference, before the next parent node is eliminated, the newly obtained compressed column should be decompressed firstly. Then, the obtained decompressed column is compressed accordingly.

In summary, the existing algorithms have the following two defects:

1) By using each phrase in the repeated instances of phrases to calculate the parameters of $c\lambda_i^j$respectively, the results are not necessarily the same. Therefore, in the compression process, not all the phrases with the same composition (for example, the first "011" and the second "011" in Table 8) can be classified into the same type of phrase.
2) In the inference process, the values of some variables in $c\lambda_i^j$, which will be obtained after eliminating a component of system, are not in its domains or do not actually exist. These problems may make the repeated instances of phrases become a run after eliminating a parent node.

3. Improvements to existing algorithms

3.1. Improved compression algorithm

For the first defect of the existing algorithms, the improved compression algorithm is developed by adding three filter conditions into the original compression algorithm. The three filter conditions are used to determine whether the current querying phrase and the existing phrase of the accompanying dictionary are the same type of "phrase".

The first filter condition is used to determine whether $Y = rem(S^{now}, 2)$ is equal to $X = rem(S^{exist}, 2)$, where S^{now} is the current querying phrase start row number in uncompressed objective sequence (OS), S^{exist} is the existing phrase of the accompanying dictionary start row number in OS and $rem(x, y)$ is a function to calculate the remainder of x/y. If $X \neq Y$, the current querying phrase will be added to the accompanying dictionary; otherwise, the second filter condition which is used to determine whether X is equal to 0 will be judged. If $X \neq 0$, the number of instances of the existing phrase will be updated; otherwise, the third filter will be judged. The third filter condition is used to determine whether $R^{now} = OS_{S^{now}-1}$ is equal to $R^{exist} = OS_{S^{exist}-1}$, where $OS_{S^{now}-1}$ is the value of the $(S^{now} - 1)$ th row in OS and $OS_{S^{exist}-1}$ is the value of the $(S^{exist} - 1)$ th row in OS. If $R^{now} = R^{exist}$, the number of instances of the existing phrase will be updated; otherwise, the current querying phrase will be added to the accompanying dictionary. The flowchart of improved compression algorithm is shown in Fig. 3.

Different from compression algorithm, improved compression algorithm has four outputs, i.e. cOS, d, RP and S^{all}. cOS is the compressed OS and d is the accompanying dictionary of cOS. Same as the outputs of inference algorithm, the j th row of cOS is $\{run, r^j, L_r^j\}$ or $\{phrase, p^j, n_p^j\}$ and the p^j th row of d is $\{p^j, v_1^j, v_2^j, L_p^j\}$. The meanings of the above variables are the same as those of the outputs of compression algorithm in Section 2.2.1. RP is the set of RP^j $(j = 1, 2, ..., m)$, i.e. $RP = \{RP^1, RP^2, ..., RP^j, ..., RP^m\}$, where m is the number of row of cOS. If the first value of the j th row in cOS is an "*run*", this run start row number in OS is the value of RP^j. Therefore, RP^j is a scalar. However, if the first value of j th row of cOS is an "*phrase*" which is the set of all the same type of phrase, these phrases start row numbers in OS compose RP^j. Thus, RP^j is an array. S^{all} represents a set that is composed by the start row numbers of all run and phrase of cOS.

For example, the column $\Pr(Sy = 0|C_1, C_2, C_3)$ in Table 2 is compressed by using the improved compression algorithm. The compressed column $\Pr(Sy = 0|C_1, C_2, C_3)$ (denoted as $cNPT_0^{Sy}$) and its accompanying dictionary d_{1-CA}^{Sy} are shown in Tables 12 and 13, respectively. Besides, $RP^1 = [1, 3]$, $RP^2 = 5$ and $RP^3 = 8$. So $RP = \{RP^1, RP^2, RP^3\} = \{[1, 3], 5, 8\}$ and $S^{all} = [1, 3, 5, 8]$.

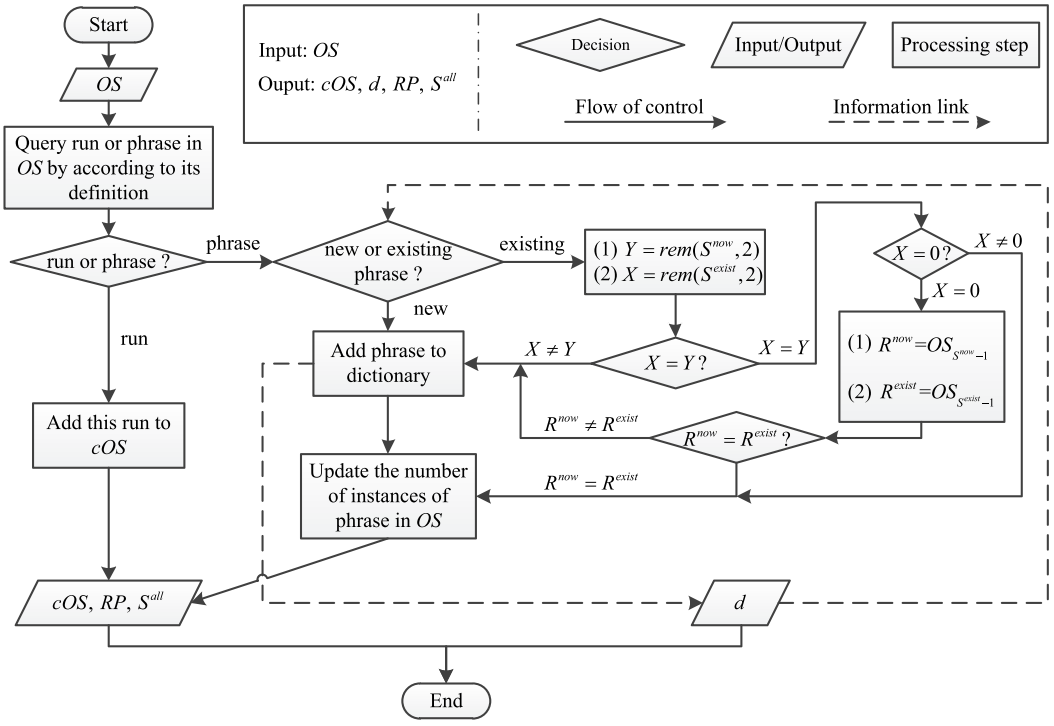


Fig. 3. Flowchart of improved compression algorithm.

Table 12
Compressed column$Sy = 0$ of Table 2.

run or phrase	r^jorp^j	L_r^j or n_p^j
phrase	1	2
phrase	2	1
run	1	1

Table 13
Accompanying dictionary of $cNPT_0^{Sy}$.

p	v_1	v_2	L_p
1	1	0	2
2	1	0	3

3.2. Improved inference algorithm

For the second defect of the existing algorithms, the improved rules for constructing $c\lambda_i{}^j$ and d_i^j are shown in Appendix E and Appendix G, where *length*(**G**) is the function for calculating the number of the elements in array **G** and *find*(**Z** = z) is the function for finding the position of z in matrix **Z**. Similar to the rows of *cOS*, the j th row of $c\lambda_i$ is $\{run, r_i^j, L_{ri}^j\}$ or $\{phrase, p_i^j, n_{p_i}^j\}$ and the p_i^j th row of d_i is $\{p_i^j, v_{1i}^j, v_{2i}^j, L_{p_i}^j\}$, where the meanings of above variables are the same as those in Section 2.2.2.

In Appendix E and Appendix G, S_{i+1}^j denotes the run or phrase start row number in λ_{i+1} defined by row $j(j = 1, 2, ..., m_{i+1})$ of $c\lambda_{i+1}$, where m_{i+1} is the number of row of $c\lambda_{i+1}$. "No existence" represents that the values of corresponding variables are absent and "No existence" is represented by "−3" in calculation process. S_{i+1}^{all} means the set of all the runs or phrases start row numbers in λ_{i+1}, RP_{i+1}^j denotes the j th element in RP_{i+1} of λ_{i+1}. For a run, J_{i+1}^j represents the position of RP_{i+1}^j in S_{i+1}^{all}. For a phrase, J_{i+1}^j represents the set of the positions of all the elements of RP_{i+1}^j in S_{i+1}^{all}. The number of elements of RP_{i+1}^j for a phrase is marked by L_{RP} and $i_{RP} = 1, 2, ..., L_{RP}$. $J_{i+1}^j(1, i_{RP})$ is the i_{RP} th element of J_{i+1}^j. $RP_{i+1}^j(1, i_{RP})$ means the i_{RP} th element of RP_{i+1}^j. R_{i+1}^j is the remainder after processing row j of $c\lambda_{i+1}$, R^{all} (defined in Appendix G) is the set of R_{i+1}^j, and $R^{all}_{J_{i+1}^j(1,i_{RP})}$ denotes the $J_{i+1}^j(1, i_{RP})$ th element of R^{all}. I refers to the position of S_{i+1}^j in S_{i+1}^{all} and R_{I-1}^{all} describes the $(I-1)$ th element of R^{all}.

For a special circumstance that a run start row number S_{i+1}^j is even and $L_{r_{i+1}}^j > 2$, the j th row and the $(j+1)$ th row of $c\lambda_i$ are constructed concurrently. Because one row of $c\lambda_{i+1}$ corresponds to two rows in $c\lambda_i$, the subsequent row numbers of j in $c\lambda_i$ should be added a 1. Besides, for this special case, the run in $c\lambda_{i+1}$ can be split equally into two equivalent runs, the first equivalent run start row number in λ_{i+1} is S_{i+1}^j and the second is $S_{i+1}^j + 1$. If $R1_{i+1}^j$ represents the start row number of the first equivalent run and $R2_{i+1}^j$ represents start row number of the second equivalent run, then $R1_{i+1}^j = S_{i+1}^j$ and $R2_{i+1}^j = S_{i+1}^j + 1$. Equivalently, $R1_{i+1}^j$ and $R2_{i+1}^j$ can be seen as the two parts which make up RP_{i+1}^j. RP_{i+1}^{update} represents the updated RP_{i+1} and S_{i+1}^{update}represents the updated S_{i+1}^{all}. Therefore, if the first value of the j th row of $c\lambda_{i+1}$ is a run of which start row number S_{i+1}^j is even and $L_{r_{i+1}}^j > 2$, after processing row j of $c\lambda_{i+1}$, both RP_{i+1} and S_{i+1}^{all} are updated to be Eqs. (6) and (7), respectively.

$$RP_{i+1}^{update} = \{RP_{i+1}^1, RP_{i+1}^2, ..., RP_{i+1}^j, ..., RP_{i+1}^{m_{i+1}}\} = \{RP_{i+1}^1, RP_{i+1}^2, ..., RP_{i+1}^{j-1}, R1_{i+1}^j, R2_{i+1}^j, RP_{i+1}^{j+1} ..., RP_{i+1}^{m_{i+1}}\} \quad (6)$$

$$S_{i+1}^{update} = \{S_{i+1}^1, S_{i+1}^2, ..., S_{i+1}^{j-1}, S_{i+1}^j, S_{i+1}^{j+1}, ..., S_{i+1}^{m_{i+1}}\} = \{S_{i+1}^1, S_{i+1}^2, ..., S_{i+1}^{j-1}, S_{i+1}^j, (S_{i+1}^j + 1), S_{i+1}^{j+1}, ..., S_{i+1}^{m_{i+1}}\} \quad (7)$$

Similarly, if the above special circumstance occurs in subsequent processing the $(j+1)$ th row, $(j+2)$ th row,$\cdots$, m_{i+1} th row of $c\lambda_{i+1}$, both RP_{i+1}^{update} and S_{i+1}^{update} are updated continually by following the above process.

In order to avoid the situation which is described in the second defect, improved inference algorithm is developed and its pseudo code is shown in Appendix F. In improved inference algorithm, when a parent node C_i is eliminated in the inference process, the newly obtained $c\lambda_i^{new}$ of the child node is decompressed to get λ_i^{new} by using RP_{i+1}^{update} and S_{i+1}^{update}, and then $c\lambda_i$ is gotten by compressing λ_i^{new} accordingly.

In Appendix F, d_i^{new} is the newly obtained accompanying dictionary of $c\lambda_i^{new}$ and λ_i^{new} is the decompressed and reorganized NPT of $c\lambda_i^{new}$. $c\lambda_i^{new-j_1}$ represents the j_1 th row of $c\lambda_i^{new}$ and its value is equal to $\{run, r_i^{new-j_1}, L_{r_i}^{new-j_1}\}$ or $\{phrase, p_i^{new-j_1}, n_{p_i}^{new-j_1}\}$, while $d_i^{new-j_1}$ represents the $p_i^{new-j_1}$ th row of d_i^{new} and its value is equal to $\{p_i^{new-j_1}, v_{1_i}^{new-j_1}, v_{2_i}^{new-j_1}, L_{p_i}^{new-j_1}\}$. In particular, all the nodes in Q must be ordered to the extreme left of NPT before starting to eliminate the parent nodes of *Ch*. The process of adjusting the order of nodes in the NPT can be done in step 2 of Appendix F.

In Appendix D, *ismember*(**X, Y**) is the function for finding the position of each element of matrix **X** in matrix **Y**, *cell2mat*(**X**) is the function for converting the cell array **X** to a matrix and *ones*(*x, y*) is the function for creating a matrix whose size and all elements are $x \times y$ and 1 respectively. $\mathbf{V}_2$ is a matrix whose size and all elements are $L_p \times 1$ and $v_{2i}^{new-j_2}$ respectively. $\mathbf{V}_\lambda{}'$ is a 2×1 cell array whose the first cell is $v_{1_i}^{new-j_2}$ and the second cell is $\mathbf{V}_2$. $\mathbf{V}_\lambda$ is a matrix whose size is $L_{p_i}^{new-j_2} \times 1$. For a run, Po_λ is a scalar and $\mathbf{A}_\lambda\{Po_\lambda, 1\} = \mathbf{V}_\lambda$ represents that the Po_λ th cell of $\mathbf{A}_\lambda$ is $\mathbf{V}_\lambda$. While for a phrase, Po_λ is an array. It is supposed that the elements of Po_λ are $Po_{k_1}^\lambda$ $(k_1 = 1, 2, ..., n_{p_{i+1}}^{new-j_2})$, i.e.

$$Po_\lambda = \left[Po_1^\lambda, Po_2^\lambda, ..., Po_{k_1}^\lambda, ..., Po^\lambda_{n_{p_{i+1}}^{new-j_2}} \right] \quad (8)$$

Therefore, $\mathbf{A}_\lambda\{Po_\lambda, 1\} = \mathbf{V}_\lambda$ represents that all the Po_1^λ th cell, Po_2^λ th cell,…, $Po_{k_1}^\lambda$ th cell,…, and $Po^\lambda_{n_{p_{i+1}}^{new-j_2}}$ th cell of $\mathbf{A}_\lambda$ are $\mathbf{V}_\lambda$. Both $\mathbf{A}_\lambda^{matrix}$ and $\mathbf{A}_\lambda^{-3}$ are a matrix. It is noteworthy that "−3" means "No existence".

4. Satellite systems MBN modeling and inference algorithms

4.1. NPT of multistate node

It is supposed that the child node *Ch* Fig. 1 has $M(M \geq 3)$ states but its parent nodes only have 2 states. In the parent nodes column of NPT, the state in the L th row of NPT for parent node C_i is calculated by formula (1) in Section 2.1. Therefore, according to the logical relation between the child node *Ch* and its parent nodes, the NPT of the child node *Ch* is obtained as shown in Table 14, where $\Pr(Ch = j_3|C_1, C_2, ..., C_n)$ $(j_3 = 1, 2, ..., M)$ means the probability of the child node *Ch* in state j_3 under the condition of one kind of parent nodes state combinations. Besides, the sum of all $\Pr(Ch = j_3|C_1, C_2, ..., C_n)$ for every kind of parent nodes state combinations is equal to 1, i.e.

Table 14
The NPT of multistate child node *Ch*.

C_1	…	C_{n-1}	C_n	$\Pr(Ch\|C_1, C_2, ..., C_n)$					
				$Ch = 1$	$Ch = 2$	…	$Ch = j_3$	…	$Ch = M$
0	…	0	0	1	0	…	0	…	0
0	…	0	1	0	1	…	0	…	0
0	…	1	0	0	1	…	0	…	0
0	…	1	1	0	0	…	1	…	0
…	…	…	…	…	…	…	…	…	…
1	…	0	0	0	1	…	0	…	0
1	…	0	1	0	0	…	1	…	0
1	…	1	0	0	0	…	1	…	0
1	…	1	1	0	0	…	0	…	1

$$\sum_{j=1}^{M} \Pr(Ch = j_3 | C_1, C_2, ..., C_n) = 1. \tag{9}$$

It is noteworthy that the values of $\Pr(Ch = j_3|C_1, C_2, ..., C_n)$ corresponding to every kind of parent nodes state combinations are hypothetical in the column $\Pr(Ch|C_1, C_2, ..., C_n)$ of Table 14. While for the child nodes in real engineering systems, the values of $\Pr(Ch = j_3|C_1, C_2, ..., C_n)$ should be obtained according to the real logical relation between the child node *Ch* and its parent nodes.

4.2. ICIA for multistate child nodes

Based on improved compression algorithm and improved inference algorithm, the ICIA is formed and extended to the application of multistate child nodes with independent binary parent nodes in this section. For the state j_3, the column corresponding to $Ch = j_3$ in Table 14 is compressed by using improved compression algorithm. The inputs of improved inference algorithm are the parent nodes' failure probability $\Pr(C_i = 0)$ $(i = 1, 2, ..., n)$, the query node $Q = \varnothing$ and the evidence $E = \varnothing$, respectively. After that, the probability of the child node *Ch* in state j_3(i.e. $\Pr(Ch = j_3)$) can be gotten by using improved inference algorithm. For $j_3 = 1, 2, ..., M-1$, through repeating the above process for $M-1$ times, probabilities of the child node *Ch* in state 1, state 2, …,and state $M-1$, i.e. $\Pr(Ch = 1)$, $\Pr(Ch = 2)$,…, $\Pr(Ch = M-1)$, can be gotten respectively. Therefore, the probability of the child node *Ch* in state *M*, i.e. $\Pr(Ch = M)$, can be calculated by using the following formula (10),

$$\Pr(Ch = M) = 1 - \sum_{j=1}^{M-1} \Pr(Ch = j_3). \tag{10}$$

To summarize, the ICIA can be presented in the flowchart as shown in Fig. 4 and the pseudo code of the ICIA is shown in Appendix C.

4.3. Satellite systems MBN modeling

According to structural logic relationship of complex multistate satellite systems, the BN reliability models can be constructed by MBN with different levels so as to further improve the inference efficiency. For example, a multistate satellite system *Sys* with independent binary units is shown in Fig. 5. By using MBN, this satellite system's BN model can be constructed to be a 2-level or 3-level BN as shown in Fig. 5.

4.4. Satellite systems reliability inference

The complex multistate satellite system *Sys* with N_{unit} binary independent units has N_{state} states. Suppose that the MBN model of *Sys* has $N(N \geq 2)$ levels. In the MBN, the non-root node U_{prq} means the *r*th node of the *p*th level and it has *q* parent nodes $\{U^r_{(p-1)1}, U^r_{(p-1)2}, ..., U^r_{(p-1)q}\}$ in the $(p-1)$th level, where $p = 2, 3, ..., N$, $q = Q_{pr}$ and $r = 1, 2, ..., N_p$. N_p is the number of nodes in the *p*th level. In particular, if $p = 2$, $r = 1, 2, ..., N_{unit}$. Equivalently, every non-root node and its parent nodes can be regarded as a simple "system" BN model in MBN. For the non-root node U_{prq}, the "system" BN is shown in Fig. 6.

If $p = 2, 3, ..., (N-1)$, the probability distribution $\Pr(U_{prq})$ of U_{prq} will be inferred by using improved compression algorithm and improved inference algorithm; otherwise, if $p = N$, it will be inferred by using ICIA. For the whole MBN model of *Sys*, the MBN reliability inference algorithm is proposed to infer the probability distribution $\Pr(Sys)$ in the following.

In the inference process of $\Pr(Sys)$, the nodes in MBN are eliminated one by one from the first level to the $(N-1)$th level. For $p = 2, 3, ..., (N-1)$ and $r = 1, 2, ..., N_p$, given the inputs in formula (11),

$$\begin{cases} \{\Pr(U^r_{(p-1)1} = 0),\ \Pr(U^r_{(p-1)2} = 0),\ ...,\ \Pr(U^r_{(p-1)q} = 0)\} \\ \Pr(U_{prq} = 0 | U^r_{(p-1)1},\ U^r_{(p-1)2},\ ...,\ U^r_{(p-1)q}) \\ Q = \varnothing \\ E = \varnothing \end{cases} \tag{11}$$

based on improved compression algorithm and improved inference algorithm, the failure probability of the non-root node U_{prq} in the *p* th level, i.e. $\Pr(U_{prq} = 0)$, can be gotten by eliminating its parent nodes $\{U^r_{(p-1)1}, U^r_{(p-1)2}, \cdots, U^r_{(p-1)q}\}$ in the $(p-1)$th level. When all the failure probabilities of the non-root nodes in the *p* th level are obtained, all the parent nodes in the $(p-1)$th level are also eliminated completely. In particular, for the complex multistate satellite system node *Sys* in the *N* th level, because the node *Sys* is multistate node, its parent nodes in the $(N-1)$th level are eliminated by ICIA so as to get the probability distribution of *Sys*. To sum up the above process, the MBN reliability inference algorithm can be presented in the flowchart as shown in Fig. 7.

5. Case study

The reliability analysis of an simple attitude determination and control system, a complex multistate system and a satellite telemetry system will be analyzed to validate the proposed method in this section. It is note that the data in the following three cases are not obtained from the real products and only assumed in this paper for method validation.

5.1. Case 1

The reliability analysis of an attitude determination and control system of TT-3 satellite is used to validate the improved compression algorithm and improved inference algorithm.

5.1.1. Problem statement of case 1

A attitude determination and control system (ADCS) of TT-3 satellite has 4 binary components (denoted as C_1, C_2, C_3 and C_4, respectively). False (F) and True (T) are two different states of ADCS. "0" and "1" are two different states of binary components, i.e. failure and normal. In this case, the NPT of ADCS is shown in Table 15. The compressed column $\Pr(\text{ADCS=T}|C_1, C_2, C_3, C_4)$ is denoted as $cNPT_T$. It is supposed that all the probabilities of components in state 0 are equal to 0.066, i.e. $\Pr(C_i = 0) = 0.066$ $(i = 1, 2, 3, 4)$.

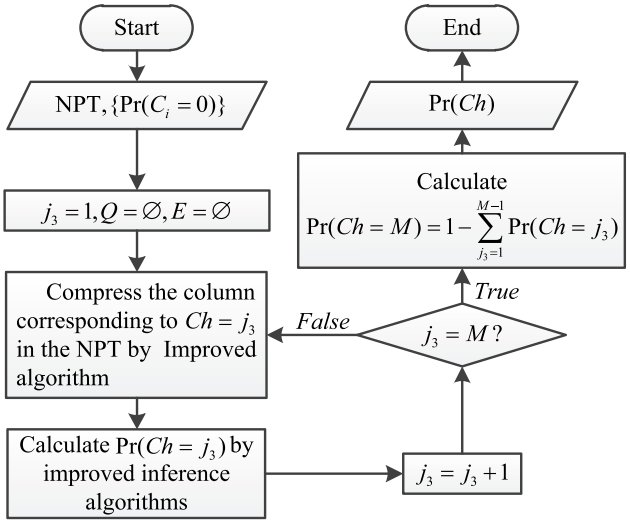


Fig. 4. Flowchart of ICIA.

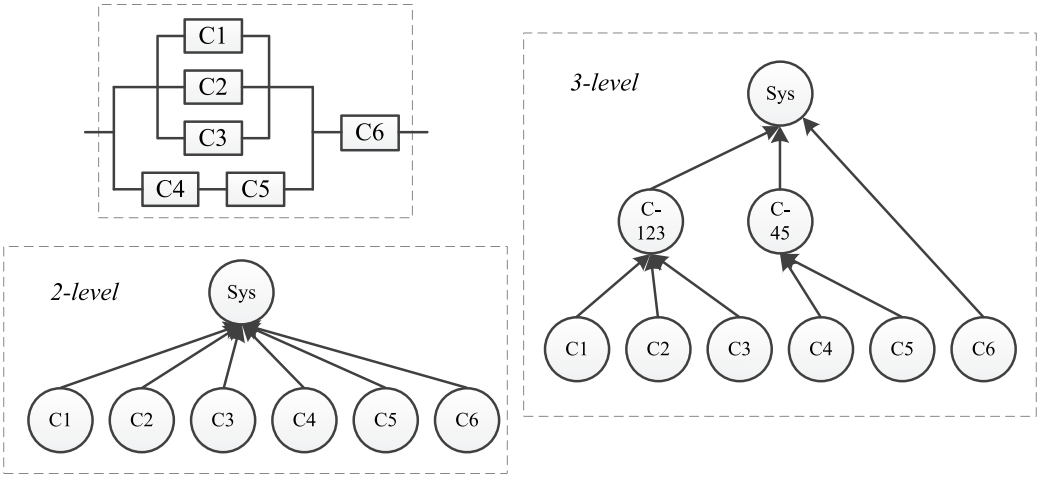


Fig. 5. A multistate satellite system and its two BN models.

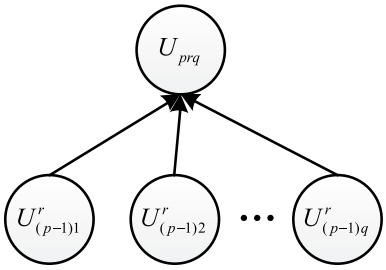


Fig. 6. "system" BN of the non-root node U_{prq}.

Table 15
NPT of ADCS.

C_1	C_2	C_3	C_4	Pr(ADCS|C_1, C_2, C_3, C_4) ADCS = F	ADCS = T
0	0	0	0	1	0
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	0	1

Table 16
$cNPT_T$ of ADCS based on the compression algorithm by Tien.

run or phrase	r^jorp^j	L^j_r or n^j_p
phrase	1	2
phrase	2	1
phrase	3	1

5.1.2. Calculation and analysis

(1) Results calculation

The column Pr(ADCS=T|C_1, C_2, C_3, C_4) is compressed by compression algorithm of Tien and the proposed improved compression algorithm, respectively. The results of compression by Tien's compression algorithm are shown in Tables 16 and 17. Besides, Tables 18 and 19 are the results of improved compression algorithm.

Based on the compressed column Pr(ADCS=T|C_1, C_2, C_3, C_4), the probability distribution of ADCS is calculated by Tien's inference algorithm and the improved inference algorithm, respectively. The results are shown in Table 20. To validate the calculation results of two methods, the probability distribution of ADCS is calculated by the benchmark AgenaRisk software [29] as shown in Fig. 8. It can be

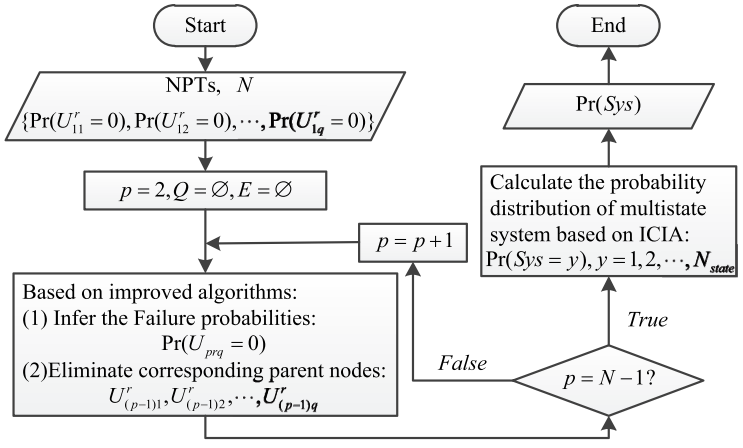


Fig. 7. Flowchart of MBN reliability inference algorithm.

Table 17
Accompanying dictionary of Table 16.

p^j	v_1^j	v_2^j	L_p^j
1	0	1	4
2	0	1	3
3	0	1	5

Table 18
$cNPT_T$ of ADCS based on the improved compression algorithm.

run or phrase	r^jorp^j	L_r^j or n_p^j
phrase	1	1
phrase	2	1
phrase	3	1
phrase	4	1

Table 19
Accompanying dictionary of Table 18.

p^j	v_1^j	v_2^j	L_p^j
1	0	1	4
2	0	1	3
3	0	1	4
4	0	1	5

Table 20
Probability distribution of ADCS calculated by two different methods.

Methods	Pr(ADCS=F)	Pr(ADCS=T)
Tien's algorithms	0.0581	0.9419
Improved algorithms	0.1078	0.8922

observed that the results of the improved algorithms proposed in this paper are exactly the same with the results of AgenaRisk software. However, the results of Tie's algorithms are different from the results of AgenaRisk software. The detailed reason for Tie's failure to correctly calculate the result will be further discussed in detail in the third part of this section.

(2) Extended analysis

With the failure probability of ADCS's component changes from 0 to 1, the true probability of ADCS is calculated by Tien's algorithms, the proposed improved algorithms and AgenaRisk software, respectively. The calculation results are shown in Fig. 9. Obviously, during the entire calculation process, the calculation result of improved algorithms is always the same with AgenaRisk software's result, but the calculation result of Tien's algorithms deviates from the benchmark result greatly. Define the deviation as the value obtained by subtracting the benchmark result from the result of Tien's algorithms. Then, the relationship between the deviation and the failure probability of component is shown in Fig. 10. Except for the values 0, 1, and 0.4 (approximate value) of the component's failure probability, all of Tien's results in other cases are biased, which indicates that the original Tien's compression and inference algorithms fail to correctly solve this reliability analysis problem.

According to the component history information of ADCS, the maximum failure probability of component usually is less than 0.2. Therefore, the ADCS's true probability and the corresponding deviation are studied when the failure probability of ADCS's component changes from 0 to 0.2 as shown in Figs. 11 and 12, respectively. According to Figs. 11 and 12, when the failure probability of component gradually increases from 0 to 0.2, the deviation is also gradually increased and the maximum deviation reaches about 0.07, which clearly demonstrates that the improvement of the proposed method compared to the original Tien's algorithm in terms of accuracy.

(3) Analysis of the causes of deviation

Comparing Table 16 with Table 18, the reason for Tien's error is that the two "0111"s in the column Pr(ADCS=T|C_1, C_2, C_3, C_4) of Table 15 are mistakenly classified into the same type of phrase during the compression of the NPT. The start row number 1 of the first "0111" is odd and the start row number 8 of the second "0111" is even. Although the two "0111"s have the same composition, the parity of two start row number are different. Therefore, the two "0111"s should not be classified into the same type of phrase.

In summary, by comparing and analyzing the calculation results of the three methods, namely the original Tien's compression and inference algorithms, the proposed improved algorithms of this manuscript and the benchmark AgenaRisk software, the performance of the improved algorithms is validated.

5.2. Case 2

The reliability analysis of a complex multistate satellite system is used to validate the ICIA and the MBN reliability inference algorithm.

5.2.1. Problem statement of case 2

The complex multistate satellite system has 44 units and is divided into six subsystems according to its functions as shown in Fig. 13. The six subsystems are attitude determination and control subsystem, onboard data handing subsystem, telemetry tracking control subsystem,

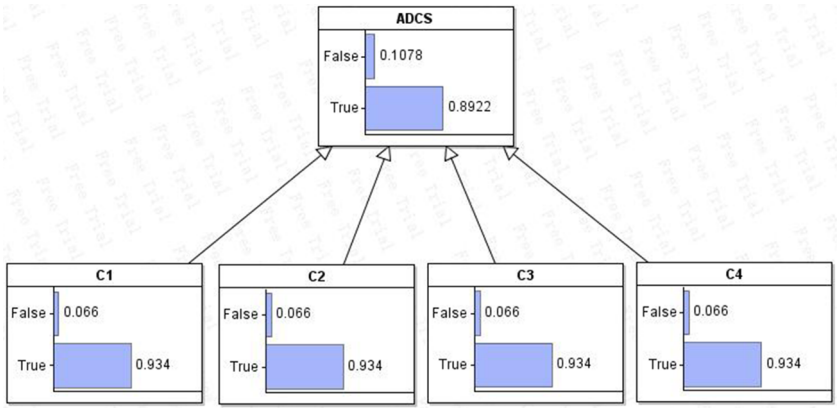


Fig. 8. Probability distribution of ADCS calculated by AgenaRisk software.

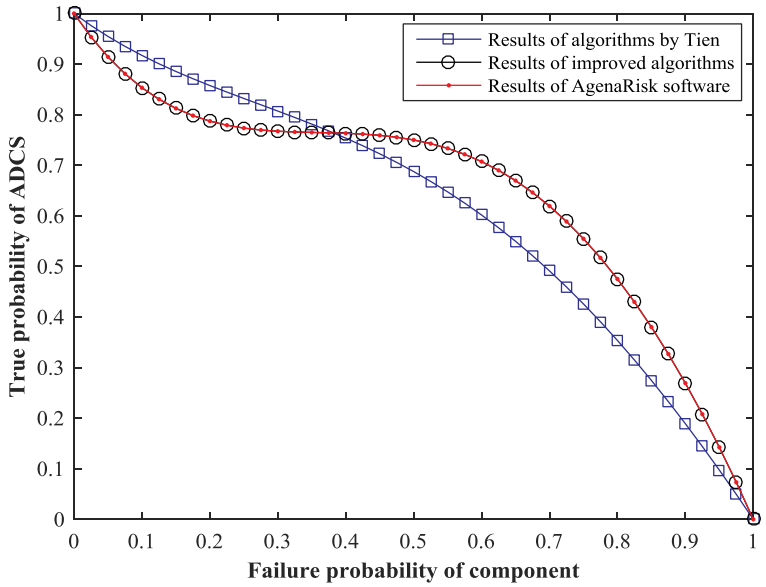


Fig. 9. True probability of ADCS when failure probability of component changes from 0 to 1.

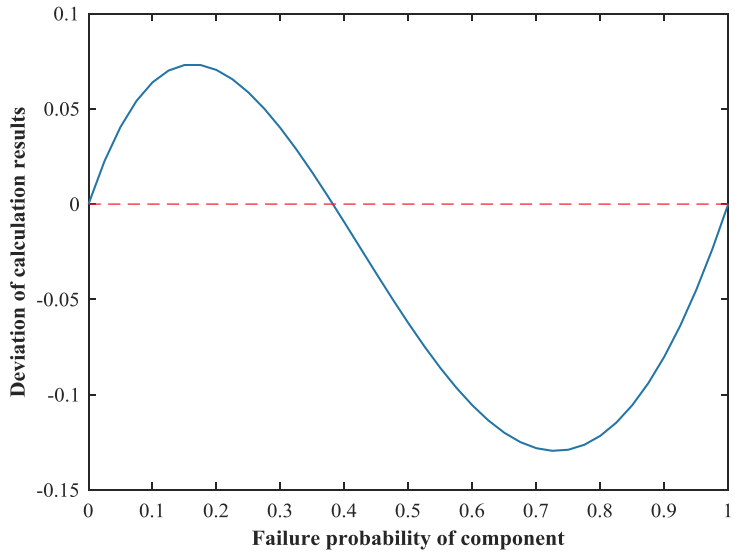


Fig. 10. Calculation deviation value when failure probability of component changes from 0 to 1.

propulsion subsystem, thermal control subsystem and power subsystem, which are denoted as A, B, C, D, E and F, respectively. Every subsystem can be divided into different parts. Take attitude determination and control subsystem A as an example, it can be divided into two parts. The first part consists of 3 parallel units (A1, A2 and A3) and the second part consists of 4 series units (A4, A5, A6 and A7). The two parts make up attitude determination and control subsystem A in parallel. All units have two states (failure or normal) and the complex multistate satellite system has four states (total failure, critical failure, minor failure and total normal). If the number of the normal subsystems is lower than 3, the state of complex multistate satellite system will be in a total failure. If any 3 subsystems of the 6 subsystems are normal, the state of complex multistate satellite system will be in a major failure. If any 4 subsystems of the 6 subsystems are normal, the state of complex multistate satellite system will be in a minor failure. If any 5 subsystems of the 6 subsystems or all the 6 subsystems are normal, the state of complex multistate satellite system will be in a normal state.

After the complex multistate satellite system runs on orbit for 180 days, the detailed failure probabilities of its units are listed in Table 21. The NPT of complex multistate satellite system will be constructed in the format of Table 14 based on the complex multistate satellite system's structural logic relationship. The goal is to analyze the reliability of this complex multistate satellite system based on the above data.

5.2.2. System reliability modeling and inference

System MBN modeling

To study the effect of different MBN structures (e.g. different levels) on the efficiency of reliability inference, the BN reliability model of complex multistate satellite system is constructed into 2-level, 3-level and 4-level by using the MBN, as shown in Fig. 14. The complex multistate satellite system node is denoted as *Sat*. Both the 3-level MBN model and the 4-level MBN model have 6 subsystems in level-2 and level-3. In addition to the 6 subsystems in level-3, it also has 12 parts in level-2, which are denoted as *A-123, A-4567, B-1234, B-567, C-1234, C-5678,D-123,D-4567, E-1234, E-567, F-1234* and *F-5678*, respectively. All the three MBN models have 44 units in level-1.

Compression NPT

The NPT of each node in three MBN models, similar to Table 1 or

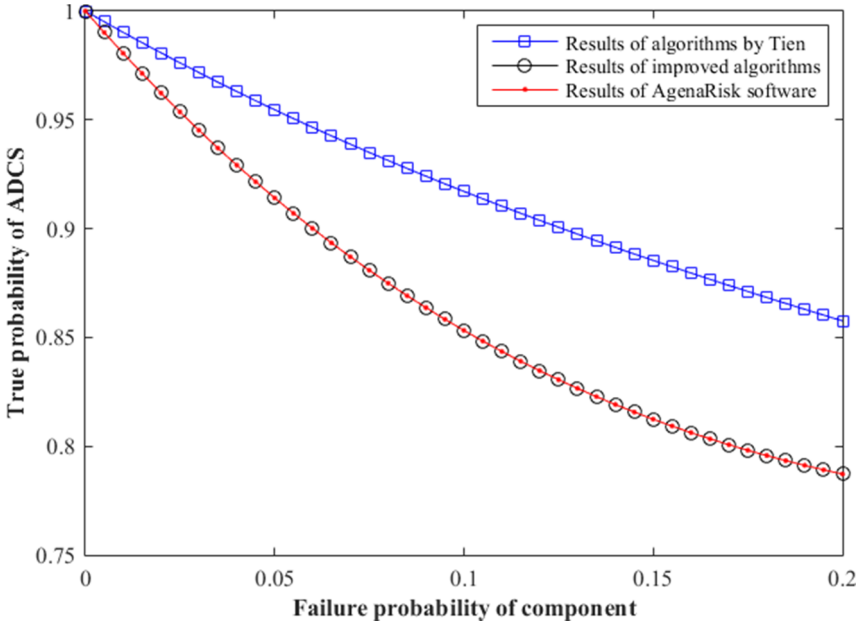


Fig. 11. True probability of ADCS when failure probability of component changes from 0 to 0.2.

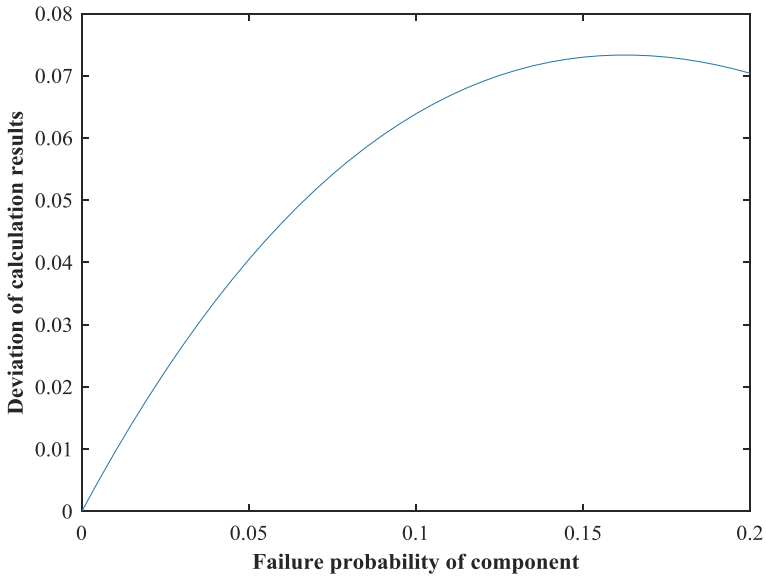


Fig. 12. Calculation deviation value when failure probability of component changes from 0 to 0.2.

Table 14, is obtained from the complex multistate satellite system's structural logic relationship. The application of improved compression algorithm is presented by taking the NPT of node *Sat* in 3-level BN of Fig. 14 as an example. "0" means the failed state of binary nodes and "1" means the normal state of binary nodes. *TF, CF, MF* and *TN* denote node *Sat* in total failure, critical failure, minor failure and total normal, respectively. Node *Sat* has 6 binary parent nodes. NPT_{Sat} represents the NPT of node *Sat* and the format of NPT_{Sat} is the same with Table 14.

In row $j_4 = 1$ of NPT_{Sat}, $\{A, B, C, D, E, F\} = \{0, 0, 0, 0, 0, 0\}$.As all parent nodes are in the failed state, the state of node *Sat* will be *To*, i.e. $\{TF, CF, MF, TN\} = \{1, 0, 0, 0\}$. In row $j_4 = 2$ of NPT_{Sat}, $\{A, B, C, D, E, F\} = \{0, 0, 0, 0, 0, 0\}$.As only one parent node is in the normal state, the state of node *Sat* will be *To*, i.e. $\{TF, CF, MF, TN\} = \{1, 0, 0, 0\}$. Similarly, checking the rest of rows ($j_4 = 3, 4, \ldots, 64$), the values of {*TF, CF, MF, TN*} can be obtained in turn. Therefore, the column $\Pr(Sat = TF|A, B, C, D, E, F)$ of NPT_{Sat} will be

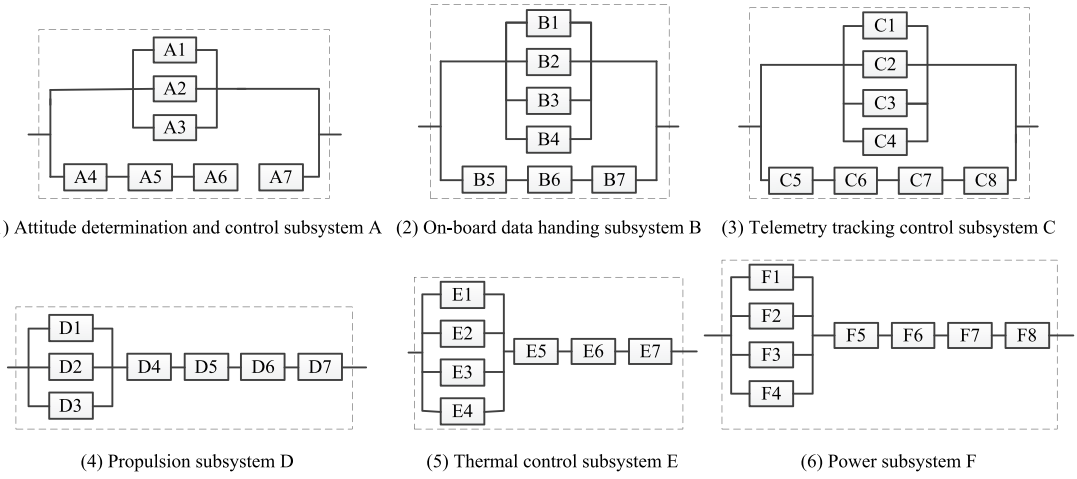
$$[1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^{T} \quad (12)$$

Then, the column $\Pr(Sat = TF|A, B, C, D, E, F)$ is compressed by using improved compression algorithm so as to get the compressed $\Pr(Sat = TF|A, B, C, D, E, F)$ (denoted as $cNPT_{TF}$) and its accompanying dictionary. The results are shown in Tables 22 and 23 respectively. The RP_{Sat} and S_{sat}^{all} are obtained as shown in Eq. (13).

$$\begin{cases} RP_{Sat} = \{1, 8, [12, 20, 36], 14, 17, 22, [25, 41], 33, 38, 49\} \\ S_{Sat}^{all} = [1, 8, 12, 14, 17, 20, 22, 25, 33, 36, 38, 41, 49] \end{cases} \quad (13)$$

Node probability inference

For 3-level MBN model of the complex multistate satellite system,



(1) Attitude determination and control subsystem A (2) On-board data handing subsystem B (3) Telemetry tracking control subsystem C

(4) Propulsion subsystem D (5) Thermal control subsystem E (6) Power subsystem F

Fig. 13. Complex multistate system's subsystems.

Table 21
Units' failure probabilities at $t = 180$ days.

Unit	P	Unit	P	Unit	P
A1	0.010	C2	0.003	E2	0.040
A2	0.020	C3	0.043	E3	0.011
A3	0.037	C4	0.028	E4	0.025
A4	0.041	C5	0.046	E5	0.045
A5	0.039	C6	0.035	E6	0.029
A6	0.016	C7	0.029	E7	0.042
A7	0.027	C8	0.041	F1	0.037
B1	0.004	D1	0.044	F2	0.029
B2	0.006	D2	0.049	F3	0.012
B3	0.007	D3	0.001	F4	0.033
B4	0.034	D4	0.043	F5	0.004
B5	0.025	D5	0.031	F6	0.031
B6	0.009	D6	0.049	F7	0.033
B7	0.025	D7	0.026	F8	0.036
C1	0.007	E1	0.024	–	–

all the failure probabilities of *A, B, C, D, E* and *F* as shown in Table 24 can be obtained after eliminating the nodes in the first level. Because *Sat* is a multistate node, the probability distribution Pr(*Sat*) of *Sat* is calculated by ICIA.

Based on Tables 22–24, the probability of *Sat* in first state *TF*, i.e.

Table 22
$cNPT_{TF}$ of NPT_{Sat}.

run or phrase	r^j or p^j	L_r^j or n_p^j
run	1	7
phrase	1	1
phrase	2	3
run	0	3
run	1	3
run	0	3
phrase	3	2
run	1	3
run	0	3
phrase	4	1

Table 23
Accompanying dictionary of Table 22.

p^j	v_1^j	v_2^j	L_p^j
1	0	1	4
2	0	1	2
3	1	0	8
4	1	0	16

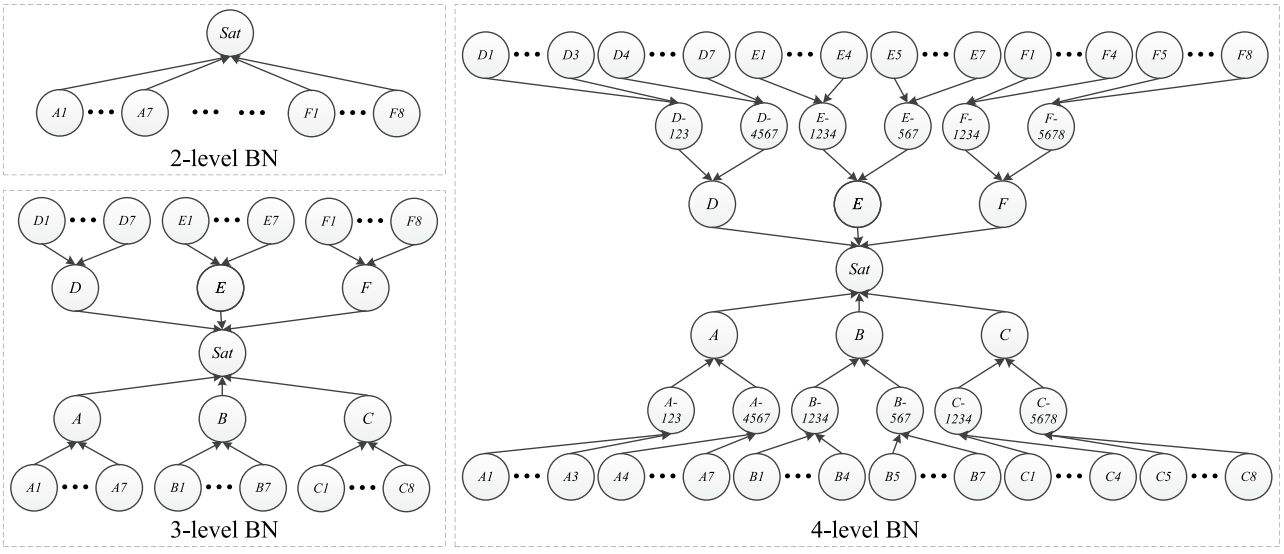


Fig. 14. Three MBN models of the complex multistate satellite system.

Table 24
Failure probabilities.

Sub	*A*	*B*	*C*	*D*	*E*	*F*
Pr(Sub = 0)	8.7047×10^{-7}	3.3090×10^{-10}	3.6090×10^{-9}	0.1410	0.1116	0.1003

$\Pr(Sat = TF)$, is obtained after eliminating *F, E, D, C, B* and *A*, successively. For example, $c\lambda_F^{new}$ and d_F^{new} are calculated as shown in Tables 25 and 26 respectively after eliminating *F*. Besides, the RP_{Sat}^{update} and S_{Sat}^{update} are obtained at the same time as shown in Eq. (14).

$$\begin{cases} RP_{Sat}^{update} = \{1, 8, [12, 20, 36], 14, 15, 17, 22, 23, [25, 41], 33, 38, 39, 49\} \\ S_{Sat}^{update} = [1, 8, 12, 14, 15, 17, 20, 22, 23, 25, 33, 36, 38, 39, 41, 49] \end{cases} \tag{14}$$

Then, taking $c\lambda_F^{new}$, d_F^{new}, RP_{Sat}^{update} and S_{Sat}^{update} as the inputs of the decompression algorithm in Appendix D, λ_F^{new} can be obtained as shown in Eq. (15).

$$\lambda_F^{new} = [1, 1, 1, 0.1003, 1, 0.1003, 0.1003, 0, 1, 0.1003, 0.1003, 0, 0.1003, 0, 0, 0, 1, 0.1003, 0.1003, 0, 0.1003, 0, 0, 0, 0.1003, 0, 0, 0, 0, 0, 0]^T \tag{15}$$

After that, taking λ_F^{new} as the input of the improved compression algorithm, $c\lambda_F$ and d_F can be gained as shown in Tables 27 and 28 respectively. Both RP_F and S_F^{all} are shown in Eq. (16) at the same time.

$$\begin{cases} RP_F = \{1, 4, 6, 8, 10, [12, 20], 14, 17, 22, 25\} \\ S_F^{all} = [1, 4, 6, 8, 10, 12, 14, 17, 20, 22, 25] \end{cases} \tag{16}$$

Similarly, $\Pr(Sat = TF)$ can be acquired after eliminating *E, D, C, B* and *A* in turn by the above process. By replacing the relevant inputs and then repeating the process of calculating, $\Pr(Sat = CF)$ and $\Pr(Sat = MF)$ can also be gotten. Besides, $\Pr(Sat = TN)$ can be calculated by formula (17).

$$\Pr(Sat = TN) = 1 - \sum_{state=TF,CF,MF} \Pr(Sat = state) \tag{17}$$

5.2.3. Results analysis

The reliability of the complex multistate satellite system is inferred by using the MBN reliability inference algorithm, the Bayes Net Toolbox (BNT) by Murphy [30] and AgenaRisk software [29], respectively. By comparing the results of these three methods, the final results of the MBN reliability inference algorithm are proved to be accurate and its inference efficiency is validated by analyzing the average calculation time. Besides, based on the MBN reliability inference algorithm method, the reliability inference is performed by using 2-level MBN model, 3-level MBN model and 4-level MBN model so as to validate the decrease of memory storage requirements on modeling BN after using improved compression algorithm and the effect of different levels of MBN on the efficiency of reliability inference.

(1) Reliability analysis

The probability distribution of complex multistate satellite system is shown in Table 29. The results of three methods are exactly the same. Thus, it is demonstrated that the proposed algorithms are correct.

(2) Inference efficiency

The calculation time of the above three methods are compared so as to validate the higher inference efficiency of the developed method, i.e. MBN reliability inference algorithm method. Both MBN reliability inference algorithm method and BNT method are run in MATLAB R2014a, and the third method is run in AgenaRisk software. All of these three methods are run on a 16 GB RAM computer with 2.5 GHz Intel(R) Core(TM) i7-4710MQ processor. The reliability of the complex multistate satellite system *Sat* is inferred by using the above three methods respectively based on 4-level MBN model. All of these three methods are run for 50 times. The average calculation time of each method is shown in Table 30. Obviously, the average calculation time of MBN reliability inference algorithm is the shortest. The inference efficiency of MBN reliability inference algorithm method is improved by 94.64% and 95.68% by comparing to BNT method and AgenaRisk software method respectively. Thus, compared with BNT method and AgenaRisk software method, the proposed MBN reliability inference algorithm method has the highest inference efficiency.

(3) Memory storage requirements

By using MBN, the BN reliability model of *Sat* is constructed to be 3-level MBN model and 4-level MBN model as shown in Fig. 14. For the uncompressed 3-level BN model, the NPT of node *Sat* has $2^6 \times 4 = 256$ cells, all the NPT of *A, B, D* and *E* have $2^7 \times 2 = 256$ cells, the NPT of both *C* and *F* have $2^8 \times 2 = 512$ cells. Thus, the overall result is $256 \times 5 + 512 \times 2 = 2304$ cells. For the uncompressed 4-level BN model, the NPT of node *Sat* has $2^6 \times 4 = 256$ cells, all NPT of *A, B, C, D, E* and *F* have $2^2 \times 2 = 8$ cells, all the NPT of $A - 123$, $B - 567$, $D - 123$ and $E - 567$ have $2^3 \times 2 = 16$ cells, all the NPT of $A - 4567$, $B - 1234$, $C - 1234$, $C - 5678$, $D - 4567$, $E - 1234$, $F - 1234$ and $F - 5678$ have $2^4 \times 2 = 32$ cells. Thus, the overall result is $256 + 8 \times 6 + 16 \times 4 + 32 \times 8 = 624$ cells. Besides, by using improved compression algorithm to compress the NPTs of MBN models, the compressed NPTs of the 3-level MBN model and the 4-level MBN model have 225 cells and 285 cells, respectively. The memory storage requirements of all MBN models are shown in Table 31.

Obviously, both the overall result of the memory storage requirements of all the compressed NPTs of 3-level MBN model and 4-level MBN model are significantly less than the overall result of all the uncompressed NPTs respectively. Thus, the overall result of the memory storage requirements of BN modeling can be reduced when using improved compression algorithm to compress the NPTs of the system BN model.

Table 25
$c\lambda_F^{new}$ after eliminating *F*.

run or phrase	r_F^{new-j} or p_F^{new-j}	L_{rF}^{new-j} or n_{pF}^{new-j}
run	1	3
phrase	1	1
phrase	2	3
run	0.1003	1
run	0	1
run	1	1
run	0.1003	1
run	0	1
phrase	3	2
run	1	1
run	0.1003	1
run	0	1
phrase	4	1

Table 26
Accompanying dictionary d_F^{new} of Table 25.

p_F^{new-j}	v_{1F}^{new-j}	v_{2F}^{new-j}	L_{pF}^{new-j}
1	0.1003	1	2
2	0.1003	−3	1
3	0.1003	0	4
4	0.1003	0	8

Table 27
Compressed column $c\lambda_F$.

run or phrase	r_F^j or p_F^j	L_{rF}^j or n_{pF}^j
run	1	3
phrase	1	1
run	0.1003	2
phrase	2	1
run	0.1003	2
phrase	3	2
run	0	3
phrase	4	1
run	0	3
phrase	5	1

Table 28
Accompanying dictionary d_F of Table 27.

p_F^j	v_{1F}^j	v_{2F}^j	L_{pF}^j
1	0.1003	1	2
2	0	1	2
3	0	0.1003	2
4	1	0.1003	3
5	0.1003	0	8

Table 29
The probability distribution of *Sat*.

State	*TF*	*CF*	*MF*	*TN*
Probability	1.3813×10^{-9}	0.0016	0.0364	0.9621

Table 30
Average calculation time of each method.

Method	Proposed algorithms	BNT	AgenaRisk
Time(s)	0.6800	12.6805	15.7280

Table 31
Memory storage requirements.

MBN		3-level	4-level
Storage	Uncompressed	2304	624
	Compressed	225	285

(4) Effects of MBN

Both the 3-level MBN model and 4-level MBN model run 50 times by using the MBN reliability inference algorithm method. The calculation time of the two models are shown in Fig. 15. The average calculation time of 3-level MBN and 4-level MBN are 0.81 s and 0.68 s, respectively. In Fig. 15, the time required for each calculation of 4-level MBN model is smaller than the time required for each calculation of 3-level MBN model.

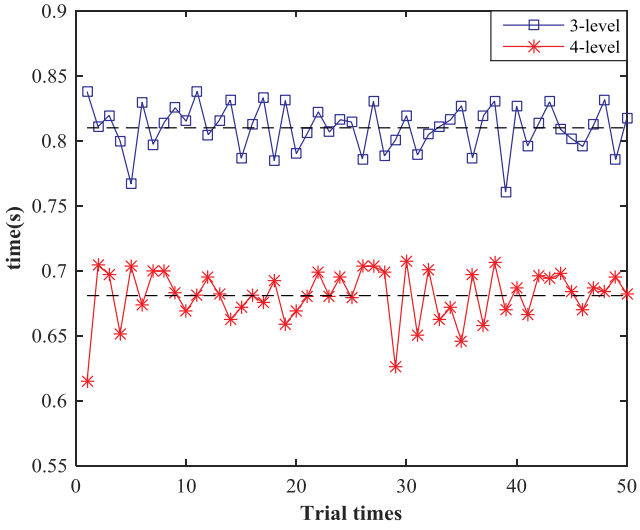


Fig. 15. Calculation time.

Compared with the 3-level MBN model, although the sum of the memory storage requirements of all the compressed NPTs of the 4-level MBN model are larger than 3-level MBN model, the average calculation time of 4-level MBN model decreases by 16.05% than that of 3-level MBN model. Especially, when the MBN model needs to be called to perform inference multiple times, the inference efficiency of 4-level MBN model becomes more prominent. For example, it is supposed that the optimal results can be obtained after iterative optimization 100 times for the optimization of complex multistate satellite system reliability. Each optimization needs to call MBN model 500 times, i.e. the entire optimization process needs to call MBN model 25,000 times. Thus, the total calculation time of calling 3-level MBN model and 4-level MBN model are 337.5 min and 283.33 min respectively. Obviously, the time cost of optimization by using 4-level MBN model is only 83.95% of that using 3-level MBN model. Therefore, the efficiency of reliability inference of complex multistate satellite system can be further improved by constructing the BN models into MBN models.

5.3. Case 3

The satellite telemetry system includes parts of satellite and parts of ground. The parts of satellite are composed of sensor, modulator, transmitter and transmitting antenna, and the parts of ground are composed of receiving antenna, receiver, demodulator, processor and record display, both of which are connected in series and shown in Fig. 16.

In Fig. 16, satellite telemetry system only has 9 units. If the series unit numbers of satellite telemetry system are increased, the maximum memory storage requirement of satellite telemetry system's NPT will increase, too. Each time 2 serial units are added, starting from the initial 9 units of the satellite telemetry system, and then the maximum memory storage requirements of satellite telemetry system's NPT is calculated by using the proposed algorithms and BNT method. Both the proposed algorithms and BNT method are run on a 16 GB RAM computer with 2.5 GHz Intel(R) Core(TM) i7-4710MQ processor. The maximum memory storage requirement of satellite telemetry system's NPT are shown in Fig. 17.

In Fig. 17, with the increase of the series unit number in the satellite telemetry system, the increasing speed of maximum memory storage requirement of NPT in using the BNT method is faster than the proposed algorithm. When the satellite telemetry system has 29 units, the maximum memory storage requirement of NPT in using the BNT method are out of a computer's maximum memory storage. However, the maximum memory storage requirement of NPT is still kept at a low

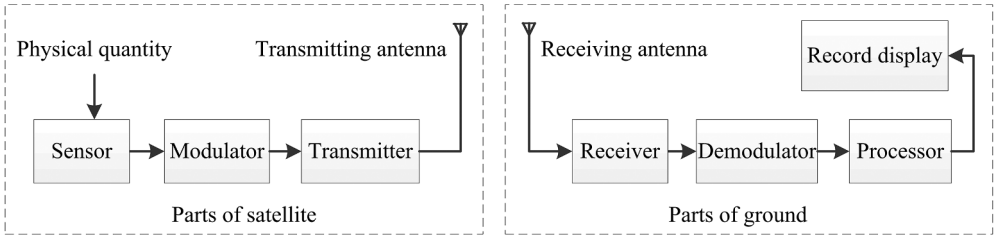


Fig. 16. Satellite telemetry system.

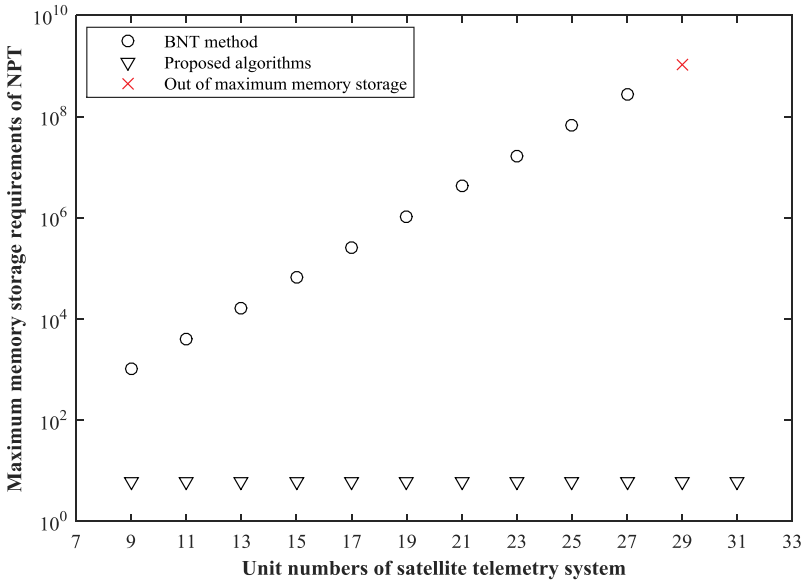


Fig. 17. Maximum memory storage requirements of satellite telemetry system's NPT.

level in using the proposed algorithm.

6. Conclusions

The proposed algorithms address the computationally unaffordable problem of the reliability analysis caused by the state combinations explosion of complex multistate satellite systems in using Bayesian Network. An improved compression algorithm effectively solves the defects of the existing compression algorithm in the inference process to make the compression techniques applicable to any NPTs of any binary nodes. Based on the compressed NPT, an improved inference algorithm effectively solves the defects of the existing inference algorithm to ensure the correctness of the inference results. Improved compression algorithm and improved inference algorithm are validated by analyzing the reliability for an attitude determination and control system of TT-3 satellite compared with the results of AgenaRisk software. Based on improved compression algorithm and improved inference algorithm, an ICIA effectively calculated the probability distributions of multistate child nodes with independent binary parent nodes by the compressed child node columns of NPTs. For complex multistate satellite systems and satellite telemetry system, a MBN reliability inference algorithm based on improved compression algorithm, improved inference algorithm, ICIA and MBN reduces the memory storage requirements of the BNs modeling and improves reliability inferences efficiency of the BN models significantly. Compared with the BNT method and the AgenaRisk software method, the higher efficiency of the MBN reliability inference algorithm method is validated by analyzing the reliability of a complex multistate satellite system. Besides, for the BN models of the complex multistate satellite systems, the application of MBN can further improve the inference efficiency, and it also makes the inferences be feasible, where the inferences originally cannot be performed due to the limitation of the maximum storage capacity of computing platform.

To sum up, the developed algorithms have four main advantages: 1) Compared with the original compression algorithm and inference algorithm, improved compression algorithm can compress the child columns of NPTs for any binary child nodes, and improved inference algorithm ensures the correctness of inference results by using the compressed child columns of NPTs; 2) ICIA effectively extends compression techniques to the applications of multistate child nodes with binary parent nodes; 3) MBN reliability inference algorithm method enables the use of BN to model and analyzes the reliability of complex multistate satellite systems, which cannot be handled with compression algorithm and inference algorithm; 4) The application of MBN model makes the inferences efficiency of BN models of complex multistate satellite systems be further improved.

Acknowledgment

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2019.04.011.

Appendix A. Rules for constructing $c\lambda_i^j$

Switch	Conditions		r_i^j or p_i^j	$L_{r_i}^j$ or $n_{p_i}^j$	R_{i+1}^j
run	$S_{i+1}^j \in$ odd	$L_{r_{i+1}}^j \in$ odd	r_{i+1}^j	$(L_{r_{i+1}}^j - 1)/2$	$r_{i+1}^j \times \Pr(C_i = 0)$
		$L_{r_{i+1}}^j \in$ even	r_{i+1}^j	$L_{r_{i+1}}^j/2$	0
	$S_{i+1}^j \in$ even	$L_{r_{i+1}}^j \in$ odd	$r_{i+1}^j \times \Pr(C_i = 1) + R^{j-1}$(also $r_i^{j+1} = r_{i+1}^j$)	1 (also $L_{r_i}^{j+1}$=$(L_{r_{i+1}}^j - 1)/2$)	0
		$L_{r_{i+1}}^j \in$ even	$r_{i+1}^j \times \Pr(C_i = 1) + R^{j-1}$(also $r_i^{j+1} = r_{i+1}^j$)	1 (also $L_{r_i}^{j+1}$=$(L_{r_{i+1}}^j - 1)/2$)	$r_{i+1}^j \times \Pr(C_i = 0)$
phrase	$S_{i+1}^j \in$ odd	$L_{p_{i+1}}^j \in$ odd	p_{i+1}^j	$n_{p_{i+1}}^j$	$v_{2_{i+1}}^j \times \Pr(C_i = 0)$
		$L_{p_{i+1}}^j \in$ even	p_{i+1}^j	$n_{p_{i+1}}^j$	0
	$S_{i+1}^j \in$ even	$L_{p_{i+1}}^j \in$ odd	p_{i+1}^j	$n_{p_{i+1}}^j$	0
		$L_{p_{i+1}}^j \in$ even	p_{i+1}^j	$n_{p_{i+1}}^j$	$v_{2_{i+1}}^j \times \Pr(C_i = 0)$

Appendix B. Rules for constructing d_i^j

Switch	Condition		$v_{1_i}^j$	$v_{2_i}^j$	$L_{p_i}^j$
phrase	$S_{i+1}^j \in$ odd	$L_{p_{i+1}}^j \in$ odd	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{p_{i+1}}^j - 3)/2] + 1$
		$L_{p_{i+1}}^j \in$ even	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{p_{i+1}}^j - 2)/2] + 1$
	$S_{i+1}^j \in$ even	$L_{p_{i+1}}^j \in$ odd	$R^{j-1} + [v_{1_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{p_{i+1}}^j - 1)/2] + 1$
		$L_{p_{i+1}}^j \in$ even	$R^{j-1} + [v_{1_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{p_{i+1}}^j - 2)/2] + 1$

Appendix C. Pseudo code of ICIA

Input	(1) NPT of multistate child node Ch(2) All failure probabilities $\Pr(C_i = 0)(i = 1, 2, \ldots, n)$
Output	The probability distribution of Ch
Step 1	$Q = \emptyset$, $E = \emptyset$
Step 2	For $j_3 \leftarrow 1$ up to $M - 1$, do the following step3-4.
Step 3	Compress the column corresponding to $Ch = j_3$ in NPT of Ch for state j_3 by improved compression algorithm;
Step 4	Calculate the probability of Ch in sate j_3 by improved inference algorithm, i.e. $\Pr(Ch = j_3)$;
Step 5	Calculate the probability of Ch in state M: $\Pr(Ch = M) = 1 - \sum_{j=1}^{M-1} \Pr(Ch = j_3)$.

Appendix D. Pseudo code of the decompression algorithm

Input	$c\lambda_i^{new}$, d_i^{new}, RP_{i+1}^{update} and S_{i+1}^{update}
Output	λ_i^{new}
Step 1	$L_{c\lambda} = length(RP_{i+1}^{update})$, $L_S = length(S_{i+1}^{update})$.
Step 2	Create a cell array$\mathbf{A}_\lambda$, whose size is $L_S \times 1$.
Step 3	For $j_2 \leftarrow 1$ to $L_{c\lambda}$, do (a) and (b)
	(a)$Po_\lambda = ismember\left(RP_{i+1}^{update-j_2}, S_{i+1}^{update}\right)$
	(b)If the first value of $c\lambda_i^{new-j_2}$ is a "*run*", do:
	If the second value of $c\lambda_i^{new-j_2}$ is a "No existence", $\mathbf{A}_\lambda\{Po_\lambda, 1\} = -3$
	If the second value of $c\lambda_i^{new-j_2}$ is not a "No existence", $\mathbf{A}_\lambda\{Po_\lambda, 1\} = r_i^{new-j_2} \times ones\left(L_{r_i}^{new-j_2}, 1\right)$
	Else if the first value of $c\lambda_i^{new-j_2}$ is a "*phrase*", do:
	If the third value of $d_i^{new-j_2}$ is a "No existence", $\mathbf{A}_\lambda\{Po_\lambda, 1\} = v_{1_i}^{new-j_2}$
	If the third value of $d_i^{new-j_2}$ is not a "No existence", $L_p = L_{p_i}^{new-j_2} - 1$, $\mathbf{V}_2 = v_{2_i}^{new-j_2} \times ones(L_p, 1)$
	$\mathbf{V}_\lambda' = \{v_{1_i}^{new-j_2}, \mathbf{V}_2\}$, $\mathbf{V}_\lambda = cell2mat(\mathbf{V}_\lambda')$, $\mathbf{A}_\lambda\{Po_\lambda, 1\} = \mathbf{V}_\lambda$
	End
	End, and output A_λ.
Step 4	$\mathbf{A}_\lambda^{matrix} = cell2mat(\mathbf{A}_\lambda)$.
Step 5	Delete "-3" of $\mathbf{A}_\lambda^{matrix}$ so as to get $\mathbf{A}_\lambda^{-3}$.
Step 6	$\lambda_i^{new} = \mathbf{A}_\lambda^{-3}$.

Appendix E. Improved rules for constructing d_i^j

Switch	Condition			$v_{1_i}^j$	$v_{2_i}^j$	$L_{P_i}^j$
phrase	$S_{i+1}^j \in$ odd	$L_{P_{i+1}}^j \in$ odd	$L_{P_{i+1}}^j = 3$	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	No existence	1
			$L_{P_{i+1}}^j > 3$	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{P_{i+1}}^j - 3)/2] + 1$
		$L_{P_{i+1}}^j \in$ even	$L_{P_{i+1}}^j = 2$	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	No existence	1
			$L_{P_{i+1}}^j > 2$	$[v_{1_{i+1}}^j \times \Pr(C_i = 0)] + [v_{2_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{P_{i+1}}^j - 2)/2] + 1$
	$S_{i+1}^j \in$ even	$L_{P_{i+1}}^j \in$ odd	$R_{i-1}^{all} + [v_{1_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{P_{i+1}}^j - 1)/2] + 1$	
		$L_{P_{i+1}}^j \in$ even	$L_{P_{i+1}}^j = 2$	$R_{i-1}^{all} + [v_{1_{i+1}}^j \times \Pr(C_i = 1)]$	No existence	1
			$L_{P_{i+1}}^j > 2$	$R_{i-1}^{all} + [v_{1_{i+1}}^j \times \Pr(C_i = 1)]$	$v_{2_{i+1}}^j$	$[(L_{P_{i+1}}^j - 2)/2] + 1$

Appendix F. Pseudo code of the improved inference algorithm

Input (1) NPT of binary child node Ch.
(2) The set of query node, i.e. Q.
(3) Failure probability of binary parent node, i.e. $\Pr(C_i = 0)(i = 1, 2, ..., n)$.
(4) The set of parent nodes' evidence, i.e. E.

Output $\Pr(Ch|Q, E)$

Step 1 For $i \leftarrow 1$ to n, do
If $C_i \in E$, update $\Pr(C_i = 0)$.
End, and output the updated $\Pr(C_i = 0)$.

Step 2 For $i \leftarrow n$ to 1, do
If $C_i \in Q$, reorder C_i to extreme left in NPT, i.e. numbered 1.
End, and output $NPT^{reorder}$.

Step 3 Compress the column $\Pr(Ch = 1|C_1, C_2, ..., C_n)$ of $NPT^{reorder}$ by using the improved compression algorithm to get cOS and then $c\lambda_{n+1} = cOS$.

Step 4 Calculate the number of nodes in Q, i.e. N_Q

Step 5 For $i \leftarrow n$ to $(N_Q + 1)$, do
Based on $c\lambda_{i+1}$:
(a) Eliminate C_i.
(b) For $j_1 \leftarrow 1$ to m_{i+1}, do
Construct $c\lambda_i^{new-j_1}$ and $d_i^{new-j_1}$ according to the case rules in Appendix E and Appendix G, update RP_{i+1} to get RP_{i+1}^{update} and update S_{i+1}^{all} to get S_{i+1}^{update}.
End, and output $c\lambda_i^{new}$, λ_i^{new}, RP_{i+1}^{update} and S_{i+1}^{update}.
(c) Decompress $c\lambda_i^{new}$ to get λ_i^{new} using the decompression algorithm in Appendix D.
(d) Compress λ_i^{new} to get $c\lambda_i$, d_i, RP_i and S_i^{all} by using the improved compression algorithm.
End, and output $c\lambda_{N_Q+1}^{new}$, $d_{N_Q+1}^{new}$, $RP_{N_Q+2}^{update}$ and $S_{N_Q+2}^{update}$.

Step 6 Decompress $c\lambda_{N_Q+1}$ to get $\Pr(Ch = 1|Q, E)$ using the decompression algorithm in Appendix D.

Step 7 $\Pr(Ch = 0|Q, E) = 1 - \Pr(Ch = 1|Q, E)$

Appendix G. Improved rules for constructing $c\lambda_i^j$

Switch	Conditions			J_{i+1}^j and I	r_i^j or p_i^j	$L_{r_i^j}$ or $n_{p_i^j}$	R_{i+1}^j	R^{all}
run	$S_{i+1}^j \in$ odd	$L_{r_{i+1}^j} \in$ odd	$L_{r_{i+1}^j} = 1$	$J_{i+1}^j = find(S_{i+1}^{all} == RP_{i+1}^j)$ $I = find(S_{i+1}^{all} == S_{i+1}^j)$	No existence	0	$r_{i+1}^j \times \Pr(C_i = 0)$	$R^{all}_{J_{i+1}^j} = R_{i+1}^j$
			$L_{r_{i+1}^j} > 1$		r_{i+1}^j	$(L_{r_{i+1}^j} - 1)/2$	$r_{i+1}^j \times \Pr(C_i = 0)$	
		$L_{r_{i+1}^j} \in$ even			r_{i+1}^j	$L_{r_{i+1}^j}/2$	0	
	$S_{i+1}^j \in$ even	$L_{r_{i+1}^j} \in$ odd	$L_{r_{i+1}^j} = 1$		$r_{i+1}^j \times \Pr(C_i = 1) + R_{i-1}^{all}$	1	0	
			$L_{r_{i+1}^j} > 1$		$r_{i+1}^j \times \Pr(C_i = 1) + R_{i-1}^{all}$(also $r_i^{j+1} = r_{i+1}^j$)	1 (also $L_{r_i^j}^{j+1}=(L_{r_{i+1}^j}-1)/2$)	0	
		$L_{r_{i+1}^j} \in$ even	$L_{r_{i+1}^j} = 2$		$r_{i+1}^j \times \Pr(C_i = 1) + R_{i-1}^{all}$	1	$r_{i+1}^j \times \Pr(C_i = 0)$	
			$L_{r_{i+1}^j} > 2$		$r_{i+1}^j \times \Pr(C_i = 1) + R_{i-1}^{all}$(also $r_i^{j+1} = r_{i+1}^j$)	1 (also $L_{r_i^j}^{j+1}=(L_{r_{i+1}^j}-2)/2$)	$r_{i+1}^j \times \Pr(C_i = 0)$	
phrase	$S_{i+1}^j \in$ odd	$L_{p_{i+1}^j} \in$ odd	$L_{p_{i+1}^j} = 3$	$L_{RP} = length(RP_{i+1}^j)$ for $i_{RP} = 1: L_{RP}$, do	p_{i+1}^j	$n_{p_{i+1}^j}$	$v_{2_{i+1}^j} \times \Pr(C_i = 0)$	$L_{RP} = length(RP_{i+1}^j)$
			$L_{p_{i+1}^j} > 3$	$J_{i+1}^j(1, i_{RP}) = find(S_{i+1}^{all} == RP_{i+1}^j(1, i_{RP}))$end	p_{i+1}^j	$n_{p_{i+1}^j}$	$v_{2_{i+1}^j} \times \Pr(C_i = 0)$	for $i_{RP} = 1: L_{RP}$, do $R^{all}_{J_{i+1}^j(1,i_{RP})} = R_{i+1}^j$ end
		$L_{p_{i+1}^j} \in$ even	$L_{p_{i+1}^j} = 2$		p_{i+1}^j	$n_{p_{i+1}^j}$	0	
			$L_{p_{i+1}^j} > 2$		p_{i+1}^j	$n_{p_{i+1}^j}$	0	
	$S_{i+1}^j \in$ even	$L_{p_{i+1}^j} \in$ odd		$I = find(S_{i+1}^{all} == S_{i+1}^j)$	p_{i+1}^j	$n_{p_{i+1}^j}$	0	
		$L_{p_{i+1}^j} \in$ even	$L_{p_{i+1}^j} = 2$		p_{i+1}^j	$n_{p_{i+1}^j}$	$v_{2_{i+1}^j} \times \Pr(C_i = 0)$	
			$L_{p_{i+1}^j} > 2$		p_{i+1}^j	$n_{p_{i+1}^j}$	$v_{2_{i+1}^j} \times \Pr(C_i = 0)$	

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